RAM is a volatile memory. ~~Temporary~~ memory

ROM " " Strong "

what is the difference between polling and Interrupt

polling : polling এ প্রসেসর সবগুলো ডিভাইস কে কিছু সময় দেয়

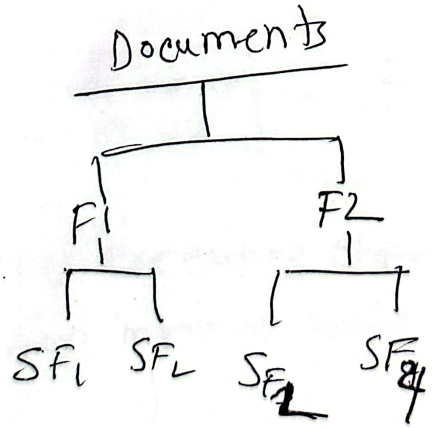
Interrupt :

permanent address: full address

relative address : related to

ls

~~ls, cd, em~~



mkdir F1

permanent address = Documents/F1/SF2

related to: F1/SF2

Terminal open

Then ls to see the folders and files.

Then comond "mkdir" to creat new Folders

The cd Foldersname

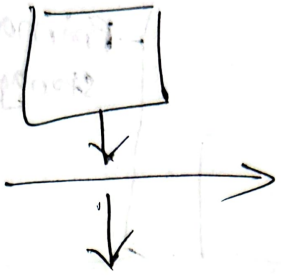
Cd .. is fora going to home.

Shortcut: for a jump SF1 to SF4

cd ~/Documents/F2/SF4

Computer System Organization

interrupt

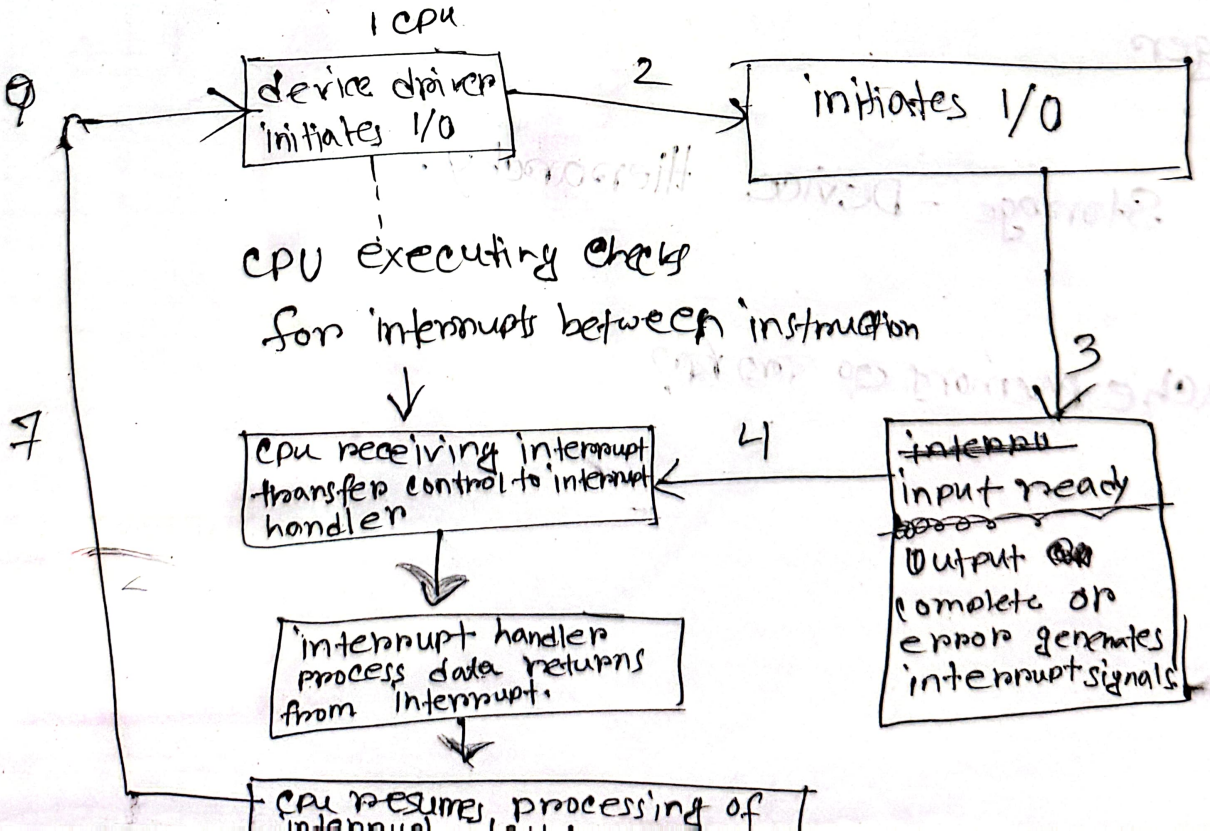


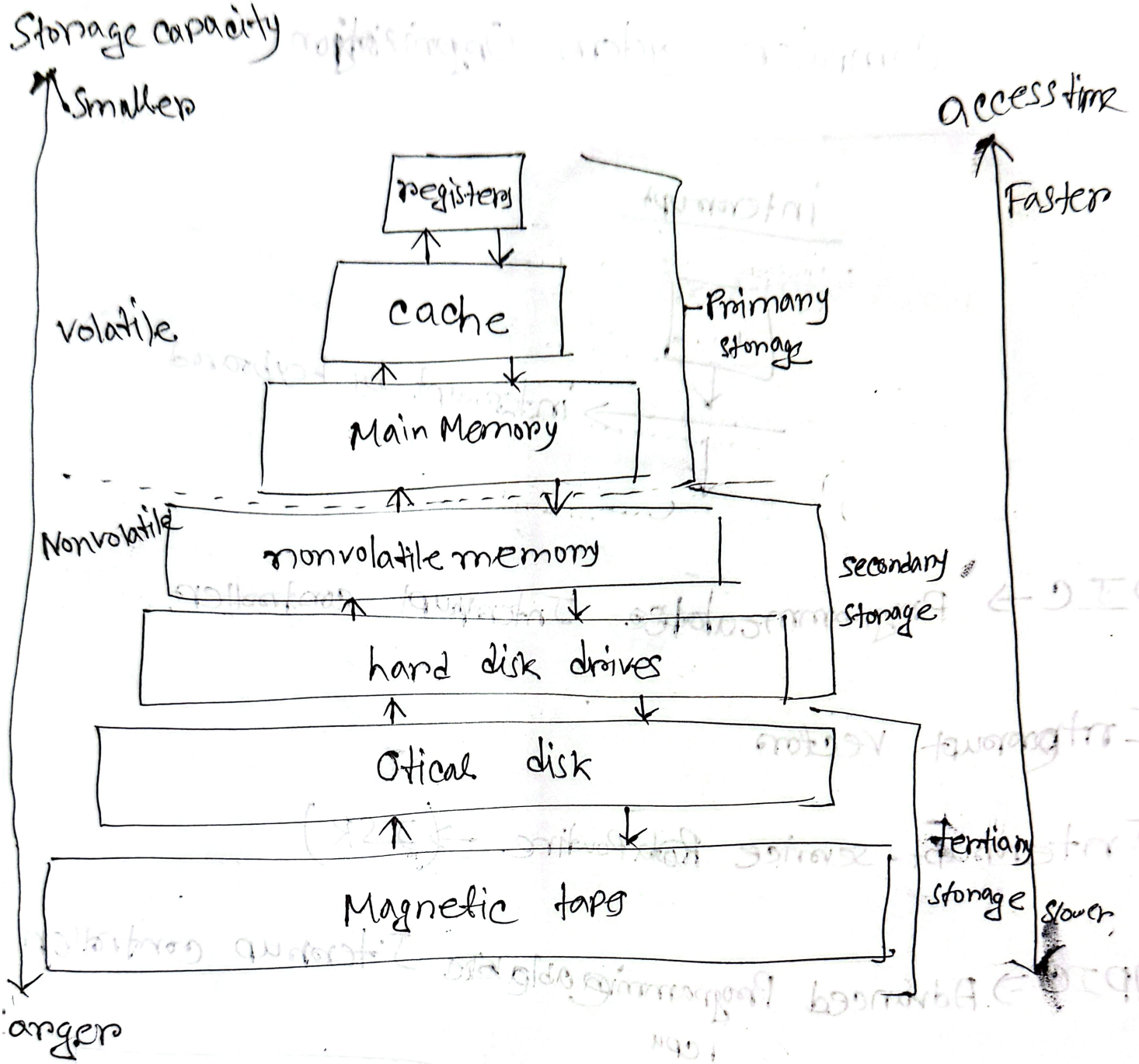
PIC → Programmable Interrupt controller.

Interrupt vectors

Interrupt service Routine. → (ISR)

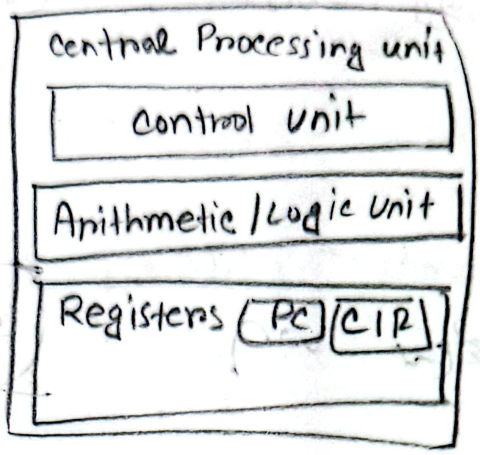
APIC ⇒ Advanced Programmable Interrupt controller





Storage - Device Hierarchy.

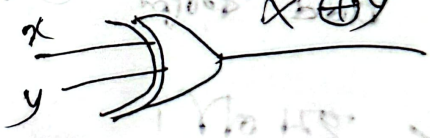
Cache memory কে বলে কি?



Number of computer

X-OR

x	y	
0	0	0
0	1	1
1	0	1
1	1	0



Lab 2

insert থেকে edit mood এসে হলে esc button

&then comments থেকে এই Shift colon (:) press করতে হবে।

Save করতে হয় w

Save + quit করতে চাইলে wq (মনে রাখবে)

grep use করে কোন text খুঁজতে হলে grep text File name

↳ it is case sensitive (case insensitive বানানোর জন্য Manual page খুলতে হবে Man grep লিখতে হবে।)

ignore case (-i)

comment এ (-i) add করলে তখন upper case or lower case এ issue হয় না।

⇒ Bandit @overthewire.:

ssh দিয়ে আমরা remote pc তে access নিতে পারি।

⇒ ssh bandit0@Domainname -P 2220 এ

কোন password দিতে হবে সেটা "bandit0" লেখা থাকে

এখন remote pc তে চলে যাব এবং কোন level এ আছি তা

cat readme দিতে হবে তাহলে পরের আমরা next level
এ যাওয়ার password পাবো।

bandit থেকে বিচ্ছিন্ন pc তে যাওয়ার জন্য exit দিতে হবে

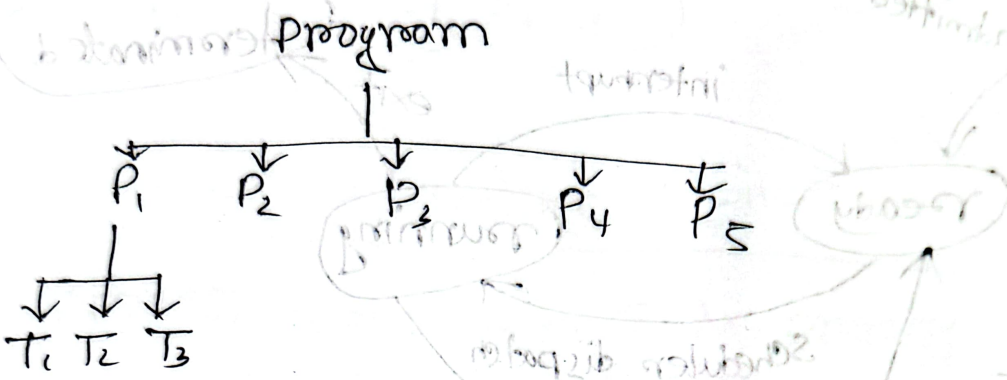
পরের level এ যেতে হলে

~~ssh bandit~~ ssh bandit1@Domainname -p 2220

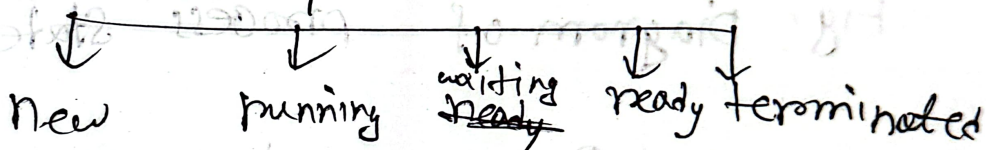
Timer

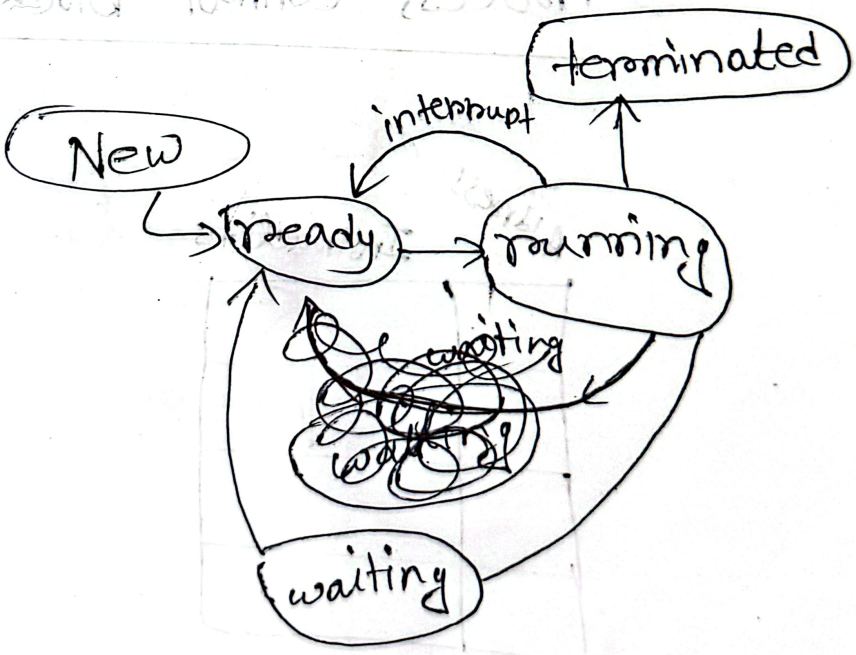
⇒ In an operating System, a timer is a hardware or software mechanism that sends regular interrupts to the cpu after a specified time interval. It enables preemptive multitasking by allowing the Operating Systems to switch between process and ensure fair cpu sharing. Additionally, it prevents programs from getting stuck in infinite loops and helps manage System time-outs.

Chapter-3 : Process



process State





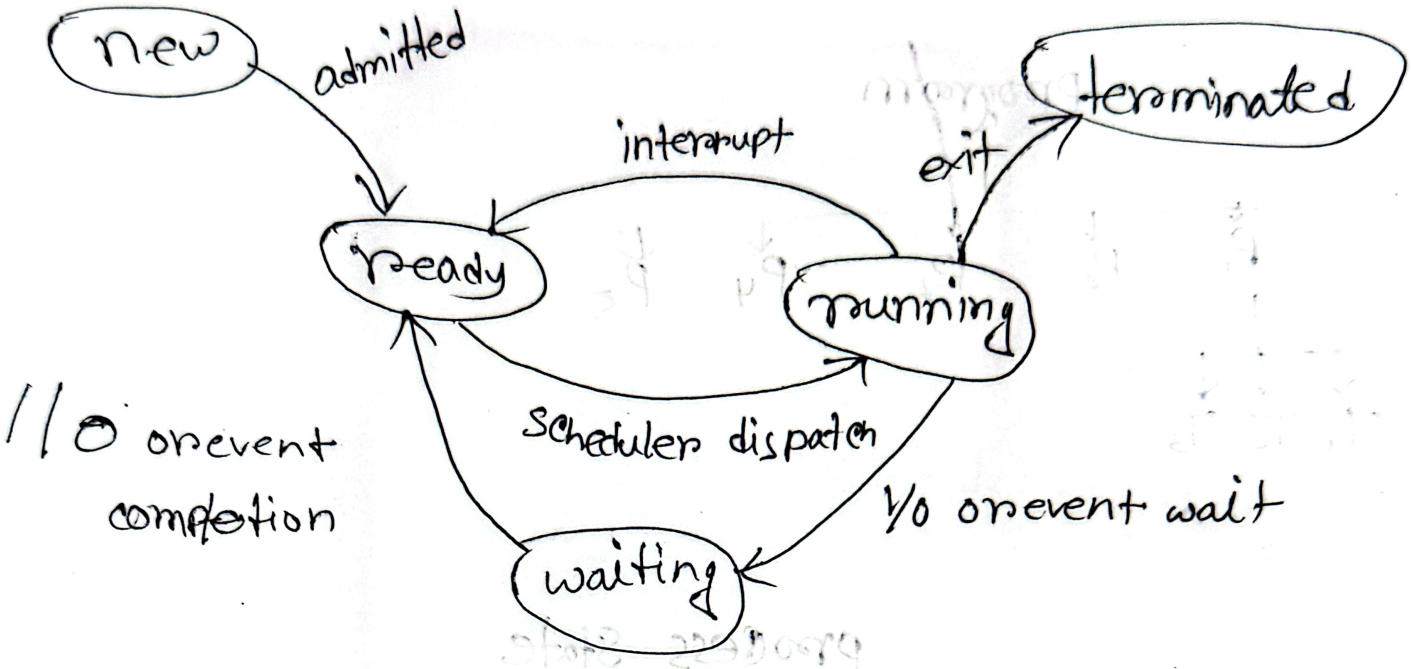
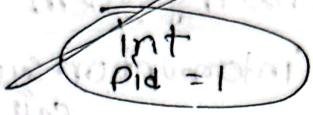


Fig: Diagram of process state.

Process Control Block (PCB)

A Trace of Process in Linux

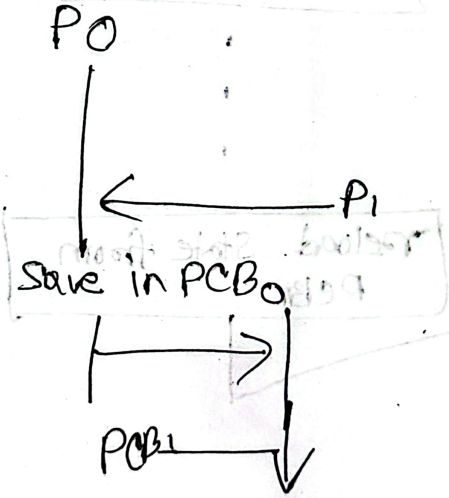


Context Switch

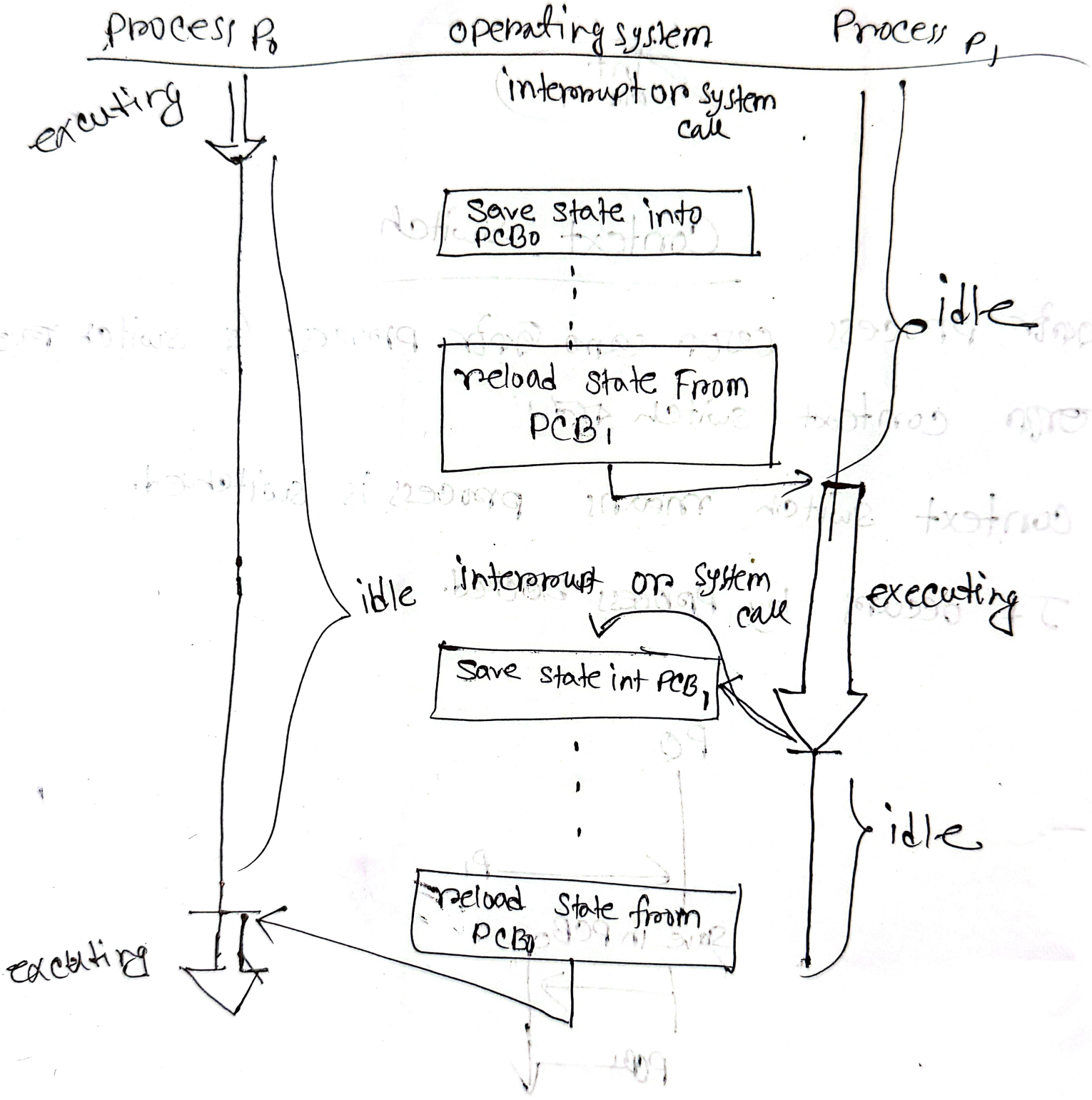
একটা process থেকে আরেকটা process এ switch করাকে বলা context switch বলে।

Context switch means process is switched.

It occurs by Process blocked.



CPU switch From process to process



Process Scheduling

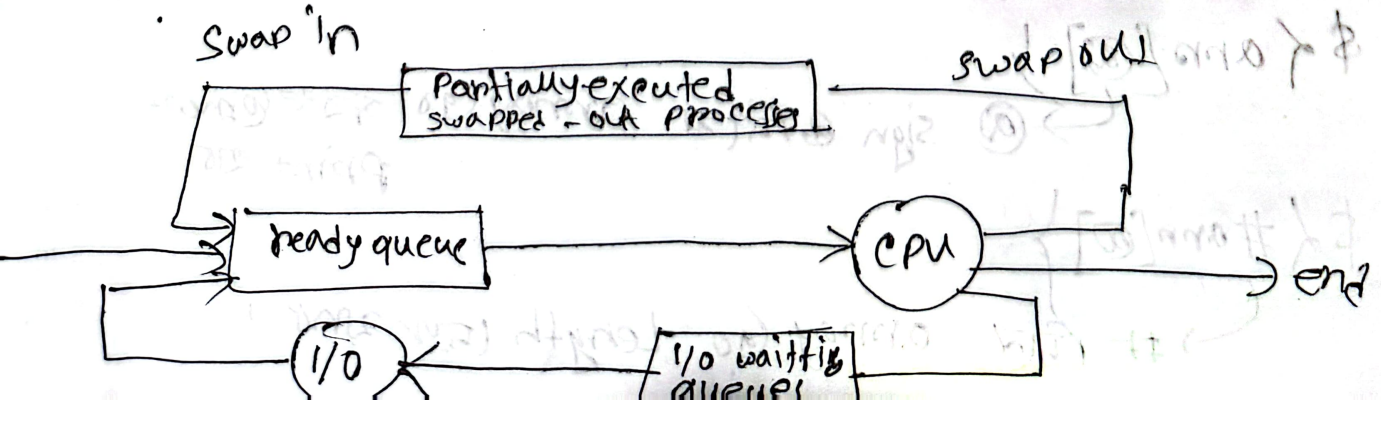
Process ~~Sheduler~~ scheduler:

scheduling queues
- → Job queue → (new is one kind of Job queue)
- → Ready queue
- → Device queue

Schedulers

Short term (CPU)	long term (Job sched

Midterm



Lab-3

vi test.sh → enter → Press i

```
#!/bin/bash
```

bash একটা language

bash এ print করার জন্য echo use করতে হয়।

```
echo "Hello world"
```

Variable print করার সময় $ sign দিতে হয়

Array

```
#!/bin/bash
arr=(5 6 7 "Hello world")
```

ফেলতে পারি / দিতে হয় না

- Sum
- Diff
- Product
- Quo
- Rem

```
echo ${arr[0]}
```

→ dollar sign

→ index number

```
echo ${arr[@]}
```

→ @ sign দিলে array এর সব element print হয়।

```
echo ${#arr[@]}
```

→ # দিলে array এর length বের হয়।

Shebang :

```
#!/bin/bash
```

Condition

```
a=10
b=20

if [ $a -lt $b ]
then
   echo " $a is smaller than $b"
fi
```

(5).

```
#!/bin/bash

number=15

if ((number%2==0));
then
    echo "$number is even."

else
    echo "$number is odd."
fi
```

```
#!/bin/bash
marks=(85 78 92 88 79)
total=0
for mark in "${marks[@]}";
  do
    total=$((total+mark))
done
```

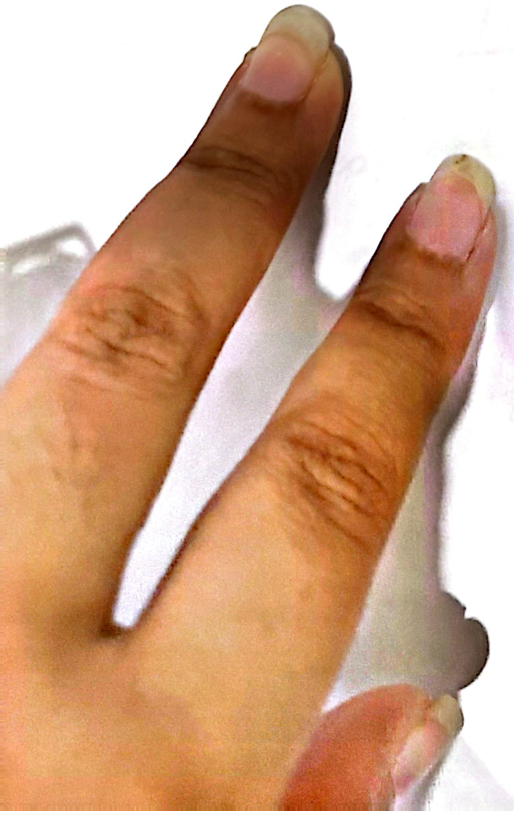
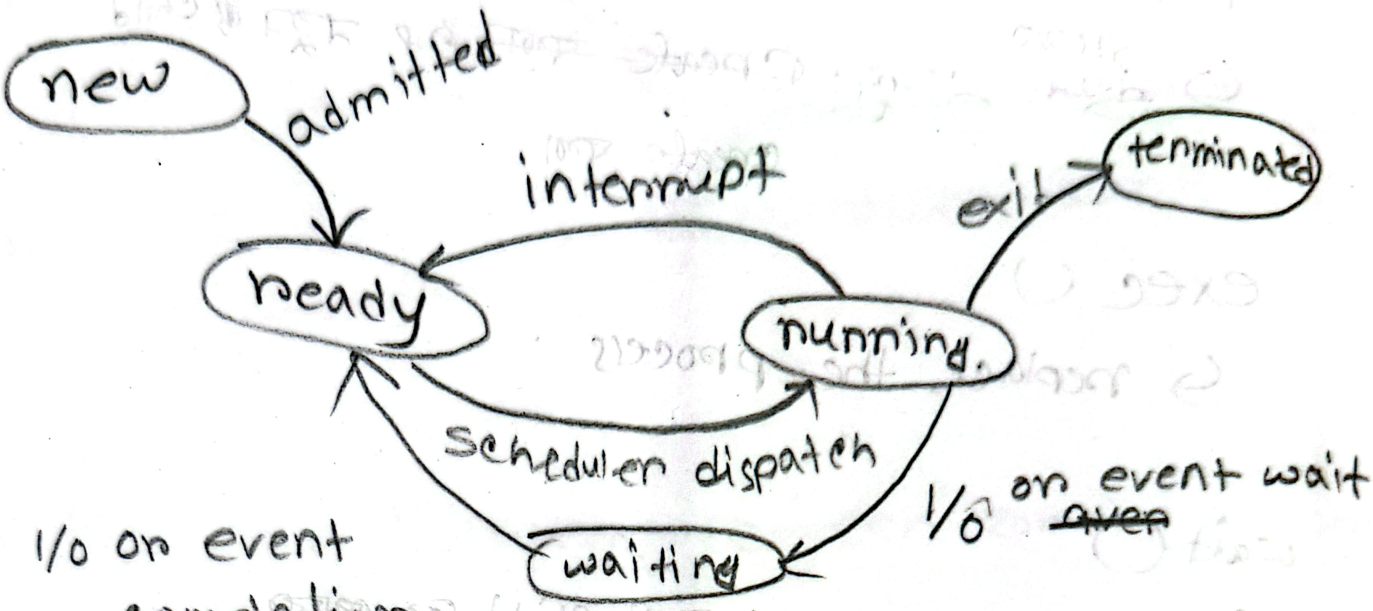
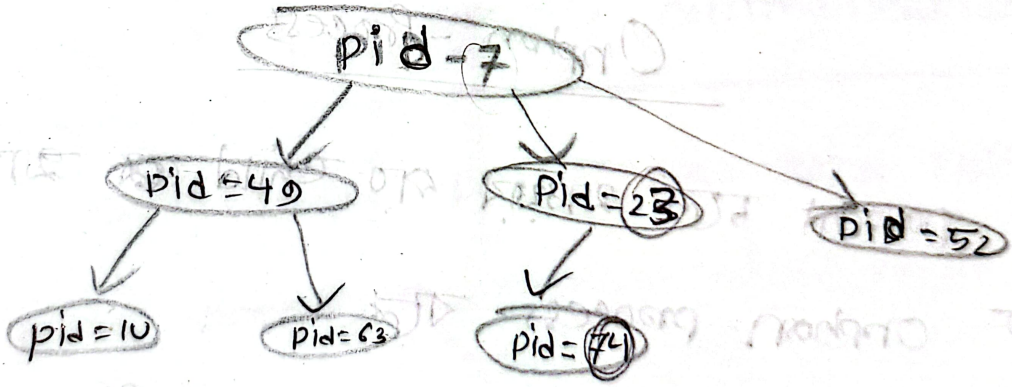


Diagram of process state:



A tree of processes in Linux



ppid of pid 74 = pid 23

ppid of pid 10 = pid 49

ppid of pid 63 = pid 49

ppid " pid 23 = pid 7

ppid " pid 49 = pid 7

ppid " pid 52 = pid 7

Fork ()

↳ প্রত্যেক ~~নতুন~~ প্রোগ্রাম ~~create করা~~ এর নতুন একটি child create করা

exec ()

↳ replace the process.

wait ()

↳ parrent report নেয় child এর থেকে (কারণ এরা child কে bye বলে দিবে)

about ()

↳ child delete করে।

Orphan - Process

যদি parent চলে যাওয়ার পর child রয়ে থাকে তাহলে তাদের orphan process বলে।

যেই child এর কোন parent থাকেনা।

Zombie process

যেসব parent process যখন wait কে call করে child কে বলে। যদি parent class wait কে call না করে তখন table থেকে pid delete হয়না। এটাই Zombie process.

Q. what is the difference between zombie and orphan process.

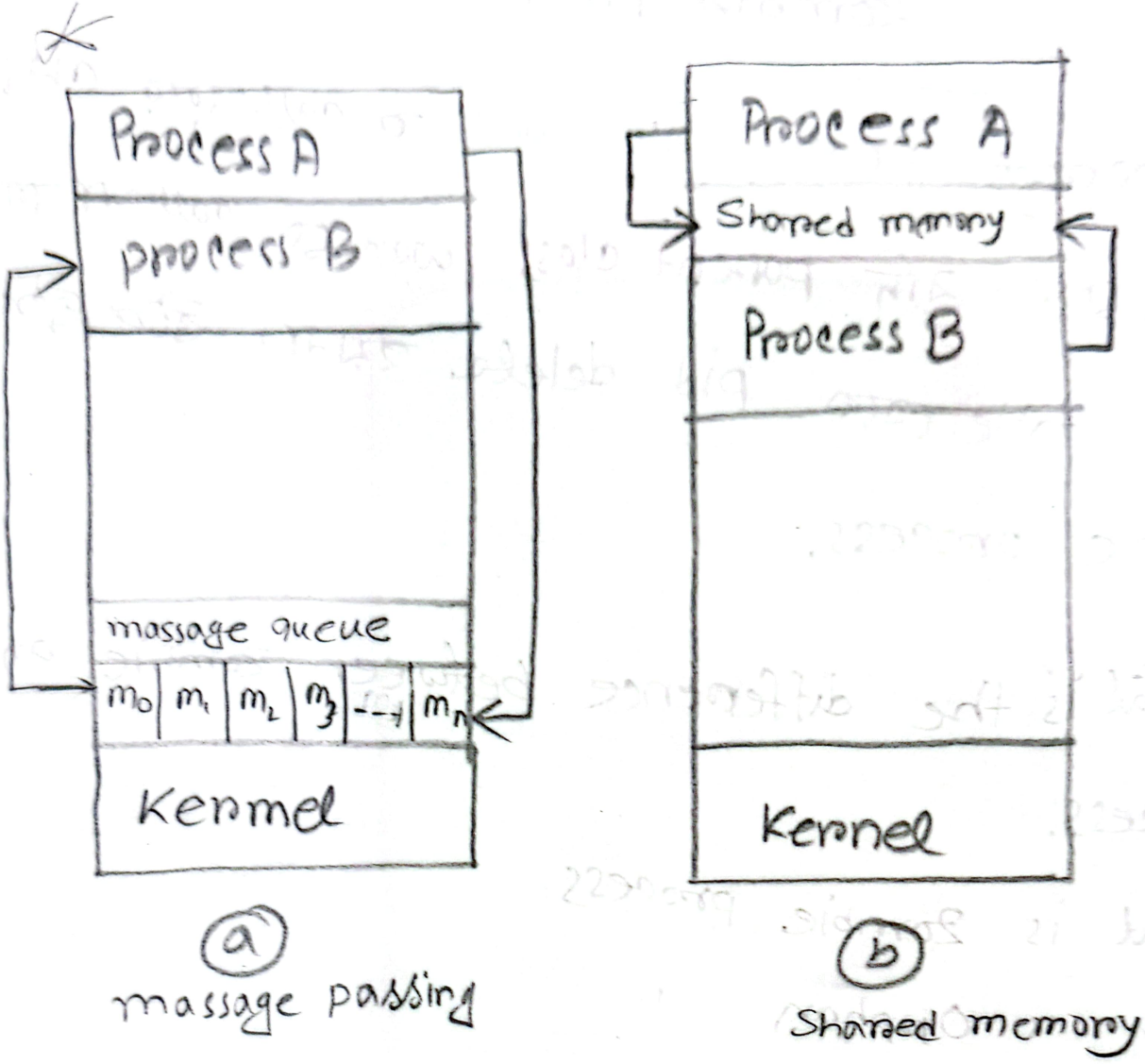
what is zombie process

" " Orphan "

Inter process communication

একটা process অন্য আরেকটি process এর সাথে যোগাযোগ করে তাকে Inter process communication বলে।

- Independent : যে সব Inter process communication এ ব্যবহার হয়না।
- Cooperating : ব্যবহার হয়।



(a) massage passing

(b) Shared memory

Producer-consumer problem:

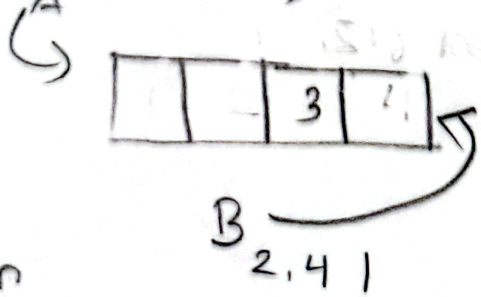
Q what is Producer-consumer Problem?

⇒

A → 1 2 3 4 5 ← B = 3, 4

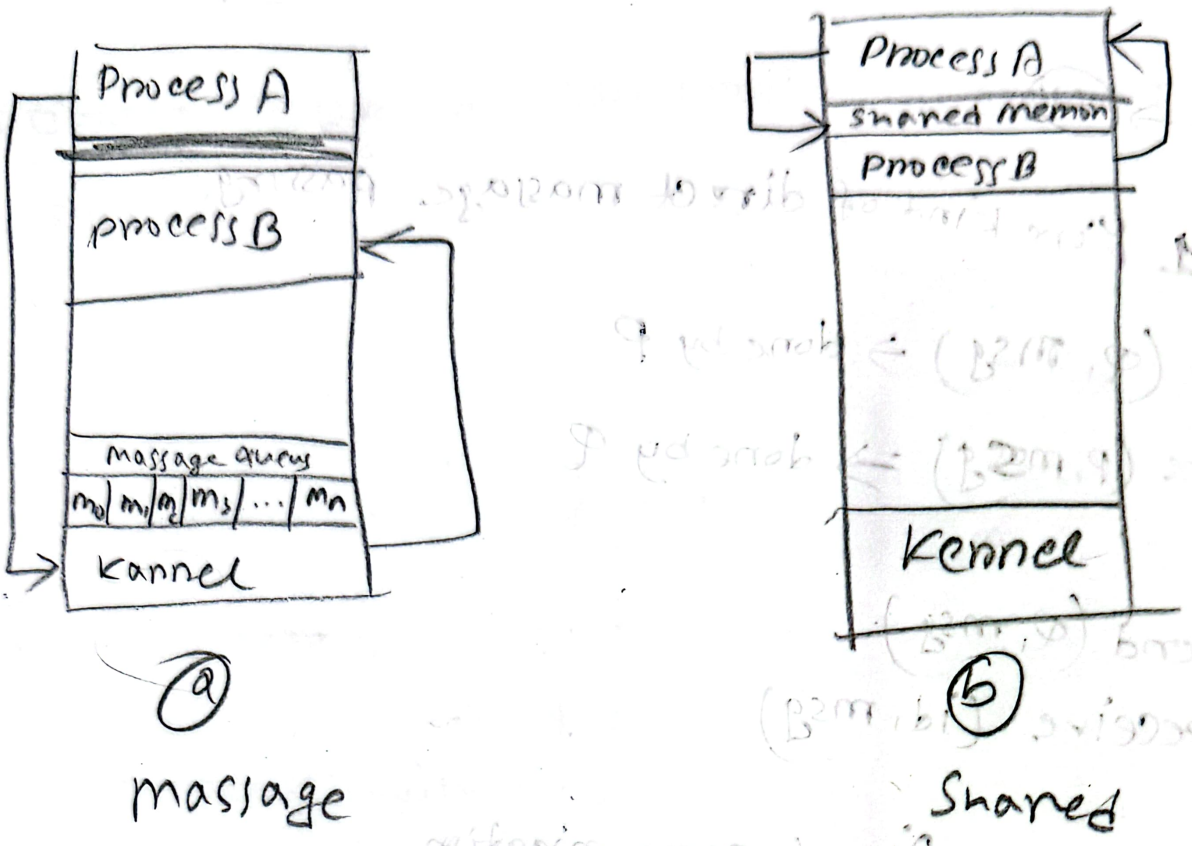
→ Unbounded buffer ⇒ size unlimited হবে

→ bounded buffer ⇒ size limited হবে।



A = producer

B = consumer

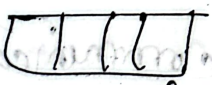


(a) message

(b) Shared

Inter process communication - message passing.

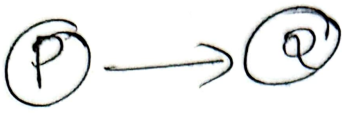
Send
receive } fixed



→ good for developer
→ bad for programmer

Variable
→ bad for computer
→ good " programmer

massage Passing (Cont.)



type 1 five kind of direct massage passing.

Sent (Q, msg) ⇒ done by P

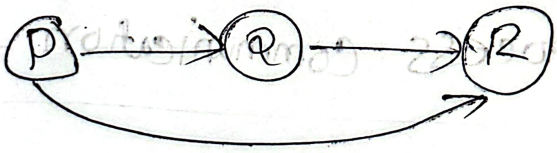
receive (P, msg) ⇒ done by Q

type 2:

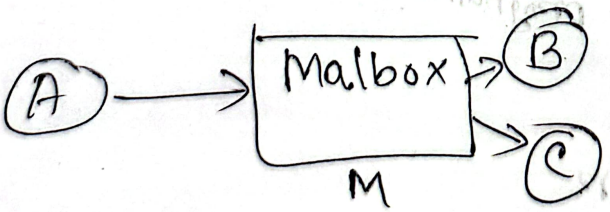
Send (Q, msg)

receive (id, msg)

Direct communication



Indirect communication



Send (M, msg)

Receive (M, Msg)

Synchronization

একজন মেসেজ পাঠানোর পর তার Receive না হওয়া পর্যন্ত সে অপেক্ষা করবে কিনা।

Buffering

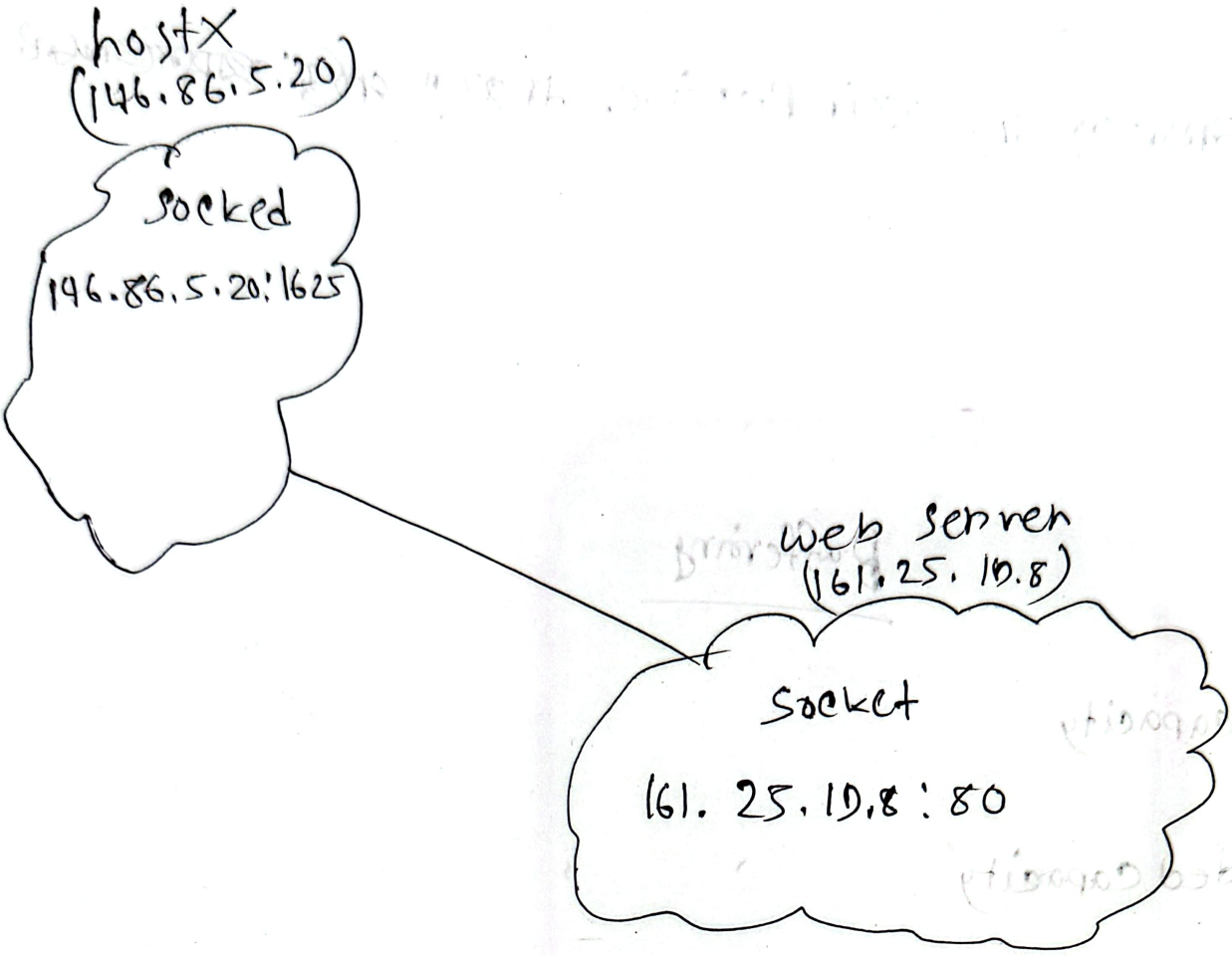
- Zero capacity
- Bounded capacity
- Unbounded capacity

Sockets

একটা ip address and code এর connection. এটাই Sockets.

well known port is fixed. 1024 is well known port. and 2220 is not well known 1024 < (....) all is not wellknown.

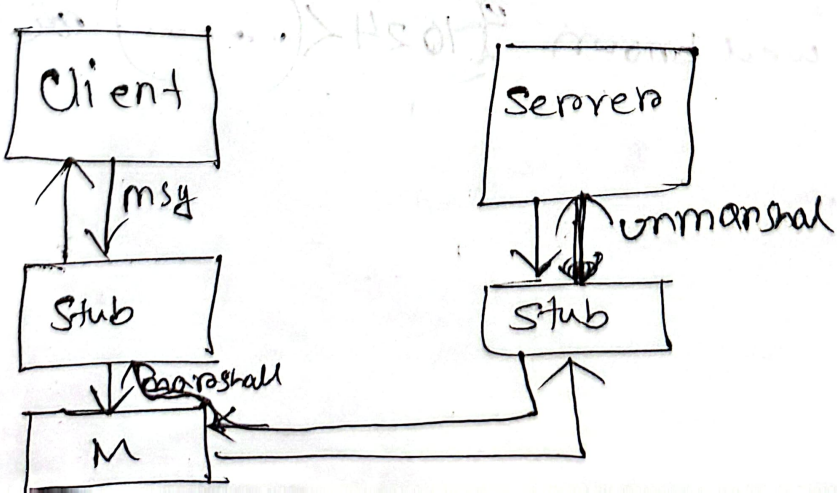
Socket communication



Remote Procedure calls

Microsoft Interface Definitation Language (MIDL)

Stubs: Client-side proxy for the actual procedure on the server.



Basic Concepts [6.4]

load program
run program } Cup Burst

↓

input ← i/o bu

↓

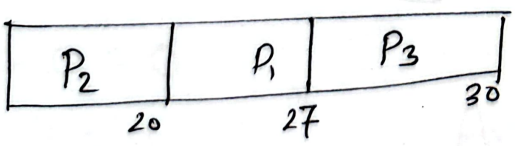
run program — Cpu B

↓

input ← i/o b

- Cpu Scheduler
- Dispatcher
- Scheduling Criteria
- Scheduling Alogo optimization Criteria
- Face Scheduling

$P_1 \rightarrow 7s$, $P_2 \rightarrow 20s$, $P_3 \rightarrow 3s$



$P_1 \Rightarrow P_2 \Rightarrow P_3$

wait of $P_2 = 0$
wait of $P_1 = 20$
wait of $P_3 = 27$

Average wait = 15.67

Basic Concepts [6.4]

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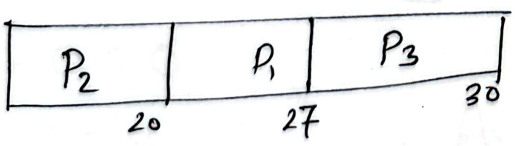


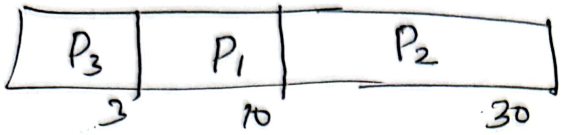
$P_1 \Rightarrow P_2 \Rightarrow P_3$

wait of $P_2 = 0$
wait of $P_1 = 20$
wait of $P_3 = 27$

Average wait = 15.67

$P_3 \rightarrow P_1 \rightarrow P_2$

wait of $P_3 = 0$
wait of $P_1 = 3$
wait of $P_2 = 10$



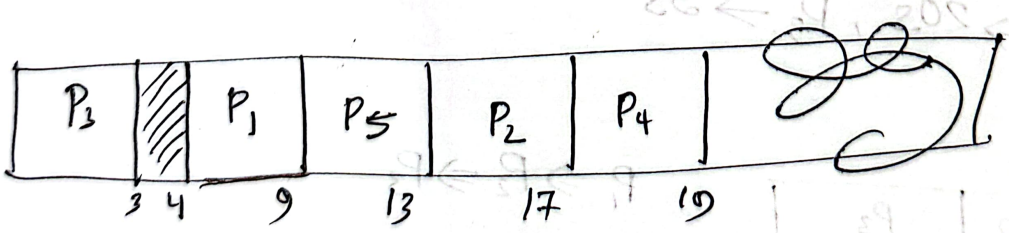
Average wait = 4.3

convoy effect

P1.

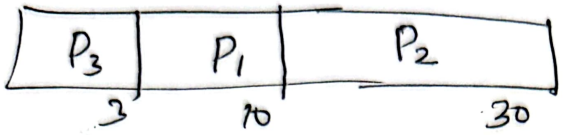
PID	Burst time	Arrival time
P_1	5	4
P_2	4	6
P_3	3	0
P_4	2	6
P_5	4	5

$P_3 \rightarrow P_1 \rightarrow P_5 \Rightarrow P_2 \rightarrow P_4$



$P_3 \rightarrow P_1 \rightarrow P_2$

wait of $P_3 = 0$
wait of $P_1 = 3$
wait of $P_2 = 10$



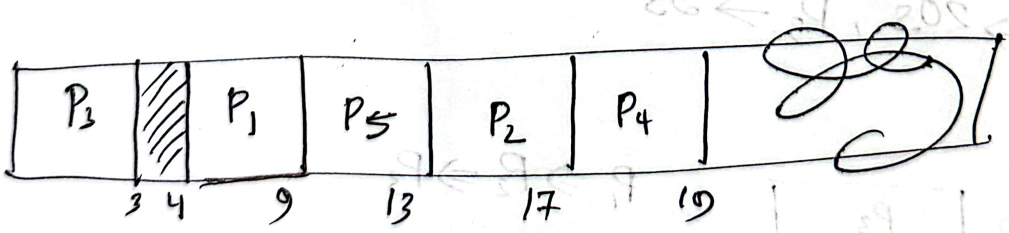
Average wait = 4.3

convoy effect

P1.

PID	Burst time	Arrival time
P_1	5	4
P_2	4	6
P_3	3	0
P_4	2	6
P_5	4	5

$P_3 \rightarrow P_1 \rightarrow P_5 \Rightarrow P_2 \rightarrow P_4$



~~Termina~~

Tarnarout Time ⇒ compilation Time - Arrival time

$P_3 = 3-0 = 3$

$P_1 = 9-4 = 5$

$P_5 = 13-5 = 8$

$P_2 = 17-6 = 11$

$P_4 = 19-6 = 13$

Ave. Tarnarount Time =

$$= \frac{40}{5} = 8$$

- waiting Time ⇒ Turnaround time - Burst time

$P_3 = 3-3 = 0$

$P_1 = 5-5 = 0$

$P_5 = 8-4 = 4$

$P_2 = 11-4 = 7$

$P_4 = 13-2 = 11$

Ave. =

$$\frac{22}{5}$$

Response time : Start of process Execution - Arrival of process

$P_3 = 0-0 = 0$

$P_1 = 4-4 = 0$

$P_5 = 9-5 = 4$

$P_2 = 13-6 = 7$

$P_4 = 17-6 = 11$

Avg Response time =

$$\frac{22}{5}$$

$= 4.4$ units

~~Termina~~

Tarnarout Time ⇒ compilation Time - Arrival time

$P_3 = 3-0 = 3$

$P_1 = 9-4 = 5$

$P_5 = 13-5 = 8$

$P_2 = 17-6 = 11$

$P_4 = 19-6 = 13$

Ave. Tarnarount Time =

$$= \frac{40}{5} = 8$$

- waiting Time ⇒ Turnaround time - Burst time

$P_3 = 3-3 = 0$

$P_1 = 5-5 = 0$

$P_5 = 8-4 = 4$

$P_2 = 11-4 = 7$

$P_4 = 13-2 = 11$

Ave. =

$$\frac{22}{5}$$

Response time : Start of process Execution - Arrival of process

$P_3 = 0-0 = 0$

$P_1 = 4-4 = 0$

$P_5 = 9-5 = 4$

$P_2 = 13-6 = 7$

$P_4 = 17-6 = 11$

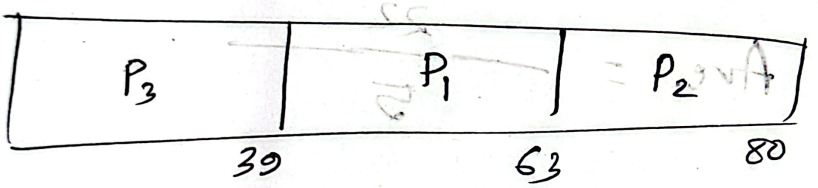
Avg Response time =

$$\frac{22}{5}$$

$= 4.4$ units

P_2.

PID	Burst Time	Response Time
P_3	24 - 15	20
P_1	17 - 26	43
P_2	39 = 22	0

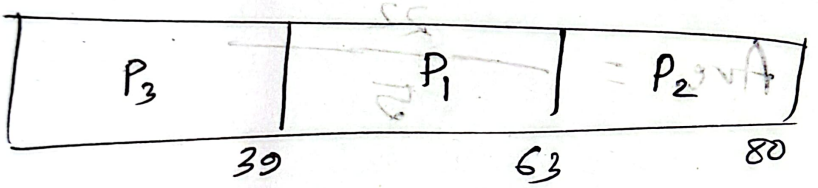


$P_3 \rightarrow P_1 \rightarrow P_2$

Turnaround Time	waiting time	Response time
$P_3 = 39 - 0 = 39$	$P_3 = 39 - 24 = 15$	$P_3 =$
$P_1 = 63 - 20 = 43$	$P_1 = 43 - 17 = 26$	$P_1 =$
$P_2 = 80 - 43 = 37$	$P_2 = 37 - 39 = -2$	$P_2 =$

P_2.

PID	Burst Time	Response Time
P_3	24 - 15	20
P_1	17 - 26	43
P_2	39 = 22	0

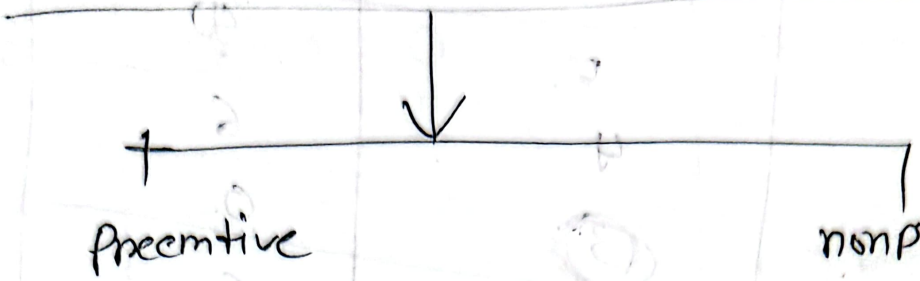


$P_3 \rightarrow P_1 \rightarrow P_2$

Turnaround Time	waiting time	Response time
$P_3 = 39 - 0 = 39$	$P_3 = 39 - 24 = 15$	$P_3 =$
$P_1 = 63 - 20 = 43$	$P_1 = 43 - 17 = 26$	$P_1 =$
$P_2 = 80 - 43 = 37$	$P_2 = 37 - 39 = -2$	$P_2 =$

SJF.

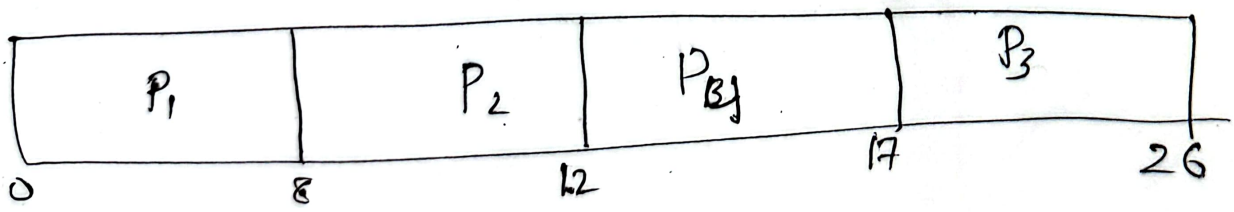
Shortest-Job-First Scheduling



SJF Algo. Simultation

PID	Arrival	Burst Time
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

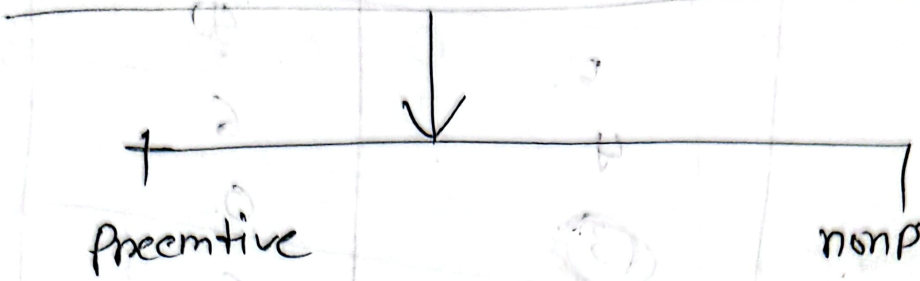
SJF Scheduling chart



Average ~~time~~ waiting =

SJF.

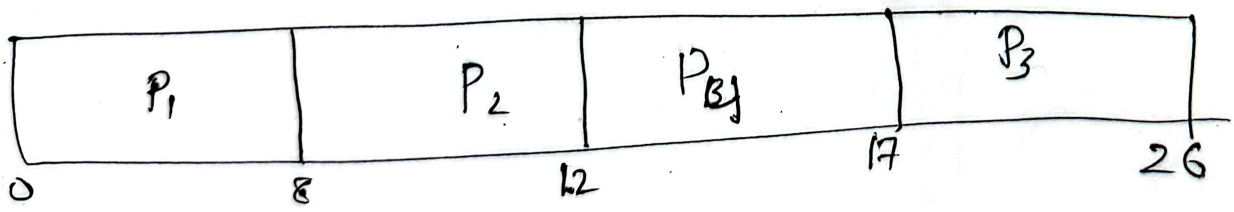
Shortest-Job-First Scheduling



SJF Algo. Simultation

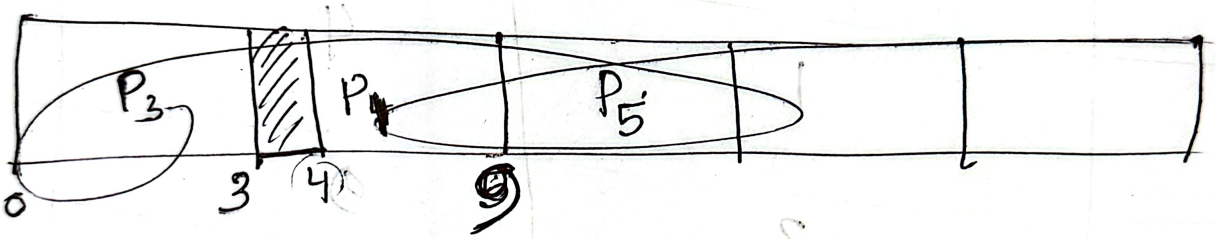
PID	Arrival	Burst Time
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

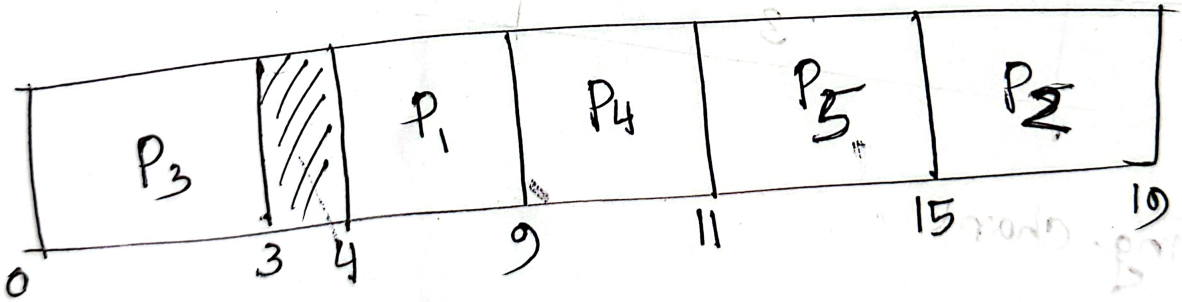
SJF Scheduling chart



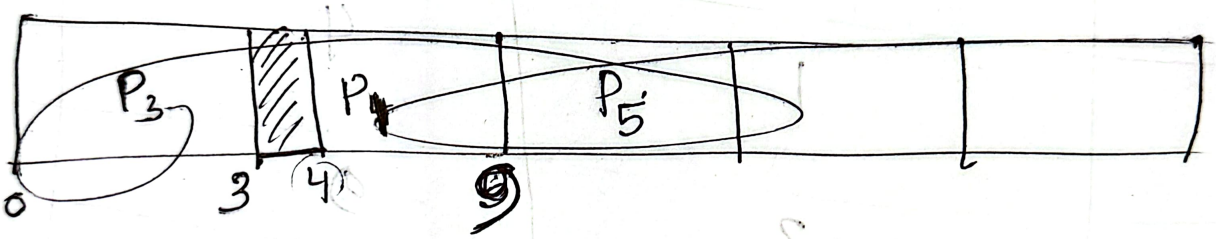
Average ~~time~~ waiting =

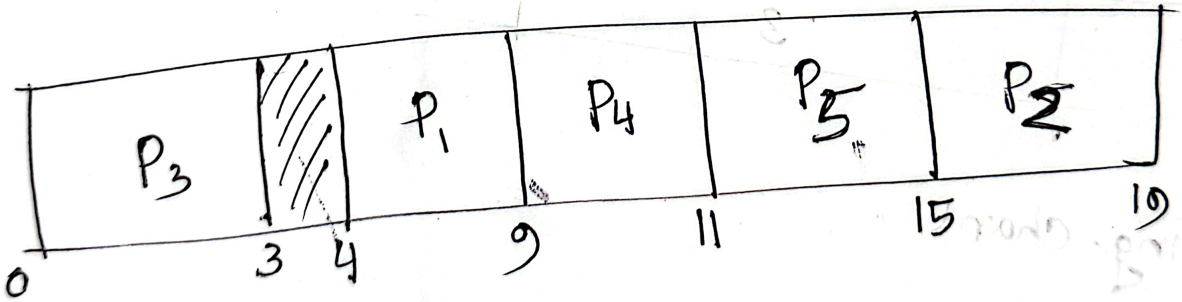
PID	Burst time	Arrival time
P_1	5.	4
P_2	4	6
P_3	3	0.
P_4	2	6
P_5	4	5





PID	Burst time	Arrival time
P_1	5.	4
P_2	4	6
P_3	3	0.
P_4	2	6
P_5	4	5





Turnaround Time ending - Arrival	waiting time = Turn - Burst =	Response time = Start time - Arrival time
$P_3 = 3 - 0 = 3$	$3 - 3 = 0$	$0 - 0 = 0$
$P_1 = 9 - 4 = 5$	$5 - 5 = 0$	$4 - 4 = 0$
$P_4 = 11 - 5 = 6$	$6 - 2 = 4$	$9 - 6 = 3$
$P_5 = 15 - 5 = 10$	$10 - 4 = 6$	$11 - 5 = 6$
$P_2 = 19 - 6 = 13$	$13 - 4 = 9$	$15 - 6 = 9$
$\frac{3+5+6+10+13}{5}$ $= 7.4$	$\frac{0+0+4+6+9}{5}$ $= 3.8$	$\frac{0+0+3+6+9}{5}$ $= 3.6$

Turnaround Time ending - Arrival	waiting time = Turn - Burst =	Response time = Start time - Arrival time
$P_3 = 3 - 0 = 3$	$3 - 3 = 0$	$0 - 0 = 0$
$P_1 = 9 - 4 = 5$	$5 - 5 = 0$	$4 - 4 = 0$
$P_4 = 11 - 5 = 6$	$6 - 2 = 4$	$9 - 6 = 3$
$P_5 = 15 - 5 = 10$	$10 - 4 = 6$	$11 - 5 = 6$
$P_2 = 19 - 6 = 13$	$13 - 4 = 9$	$15 - 6 = 9$
$\frac{3+5+6+10+13}{5}$ $= 7.4$	$\frac{0+0+4+6+9}{5}$ $= 3.8$	$\frac{0+0+3+6+9}{5}$ $= 3.6$

Shortest remaining time first

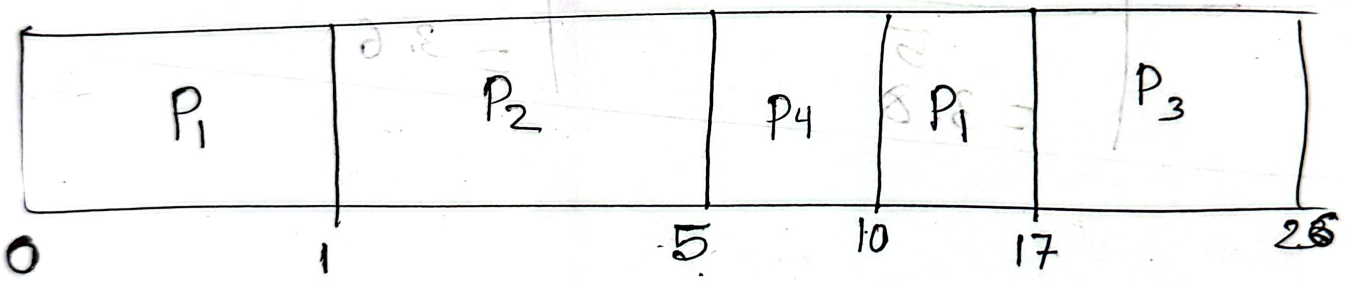
PID	Arrival	Burst
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

$P_1 \Rightarrow 8 \; 7 \; 0$

$P_2 \Rightarrow 4 \; 0$

$P_3 \Rightarrow 9 \; 0$

$P_4 \Rightarrow 5 \; 0$



Average ~~time~~ waiting time $= (9+0+15+2)/4 = 26/4$

$= 6.5$

Shortest remaining time first

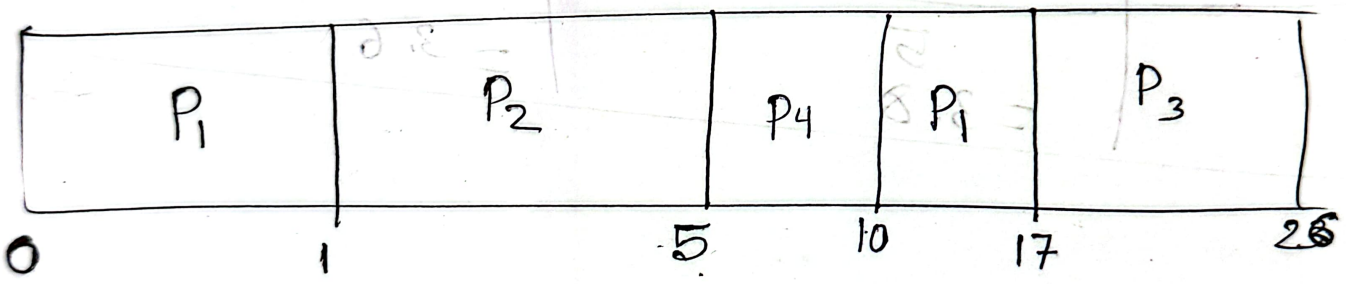
PID	Arrival	Burst
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

$P_1 \Rightarrow 8 \; 7 \; 0$

$P_2 \Rightarrow 4 \; 0$

$P_3 \Rightarrow 9 \; 0$

$P_4 \Rightarrow 5 \; 0$



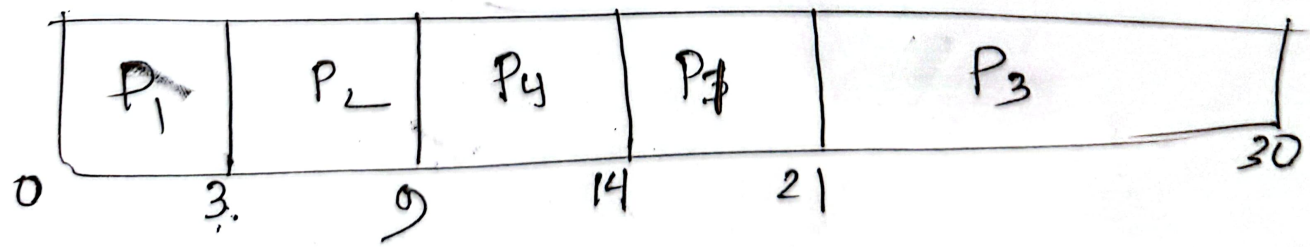
Average ~~time~~ waiting time $= (9+0+15+2)/4 = 26/4$

$= 6.5$

Turnaround Time: Completion − Burst Arrival	Waiting time: Turnaround − Burst	Response time Start − Arrival
$P_1 = 17 - 0 = 17$	$17 - 8 = 9$	$0 - 0 = 0$
$P_2 = 5 - 1 = 4$	$4 - 4 = 0$	$1 - 1 = 0$
$P_3 = 26 - 2 = 24$	$24 - 9 = 15$	$17 - 9 = 15$
$P_4 = 10 - 3 = 7$	$7 - 5 = 2$	$5 - 3 = 2$
$\frac{17 + 4 + 24 + 7}{4} = \frac{52}{4} = 13$	$\frac{9 + 0 + 15 + 2}{4} = 6.5$	$\frac{0 + 0 + 15 + 2}{4} = 4.5$

⇒

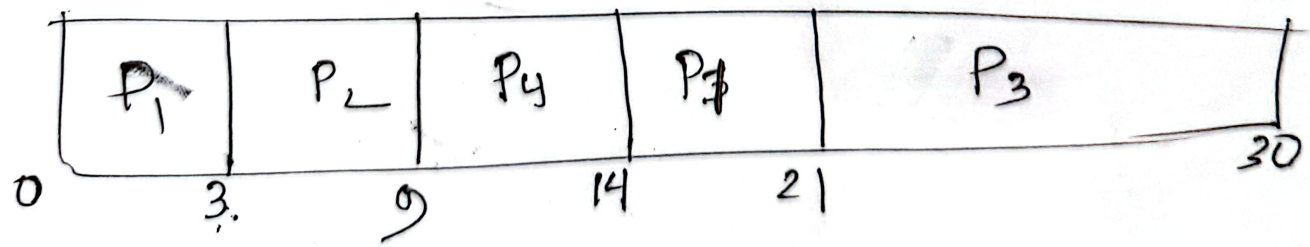
- $P_1 = 0$
- $P_2 = 0$
- $P_3 = 0$
- $P_4 = 0$



Turnaround Time: Completion − Burst Arrival	Waiting time: Turnaround − Burst	Response time Start − Arrival
$P_1 = 17 - 0 = 17$	$17 - 8 = 9$	$0 - 0 = 0$
$P_2 = 5 - 1 = 4$	$4 - 4 = 0$	$1 - 1 = 0$
$P_3 = 26 - 2 = 24$	$24 - 9 = 15$	$17 - 9 = 15$
$P_4 = 10 - 3 = 7$	$7 - 5 = 2$	$5 - 3 = 2$
$\frac{17 + 4 + 24 + 7}{4} = \frac{52}{4} = 13$	$\frac{9 + 0 + 15 + 2}{4} = 6.5$	$\frac{0 + 0 + 15 + 2}{4} = 4.5$

⇒

- $P_1 = 0$
- $P_2 = 0$
- $P_3 = 0$
- $P_4 = 0$



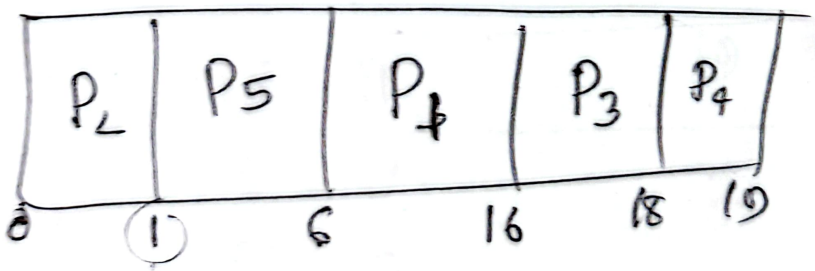
	Turnaround com - Arr	Waiting Turr. - Burst	Response Start - Arr
P_1	$= 21 - 0 = 21$	$21 - 10 = 11$	$0 - 0 = 0$
P_2	$9 - 3 = 6$	$6 - 6 = 0$	$3 - 3 = 0$
P_3	$30 - 7 = 23$	$23 - 9 = 14$	$21 - 7 = 14$
P_4	$14 - 8 = 6$	$6 - 5 = 1$	$9 - 8 = 1$
	$\frac{21 + 6 + 23 + 6}{4}$ $= 14$	$\frac{11 + 0 + 14 + 1}{4}$ $= 6.5$	$\frac{0 + 0 + 14 + 1}{4}$ $= 3.75$

	Turnaround com - Arr	Waiting Turr. - Burst	Response Start - Arr
P_1	$= 21 - 0 = 21$	$21 - 10 = 11$	$0 - 0 = 0$
P_2	$9 - 3 = 6$	$6 - 6 = 0$	$3 - 3 = 0$
P_3	$30 - 7 = 23$	$23 - 9 = 14$	$21 - 7 = 14$
P_4	$14 - 8 = 6$	$6 - 5 = 1$	$9 - 8 = 1$
	$\frac{21 + 6 + 23 + 6}{4}$ $= 14$	$\frac{11 + 0 + 14 + 1}{4}$ $= 6.5$	$\frac{0 + 0 + 14 + 1}{4}$ $= 3.75$

lab - 5

Priority Scheduling

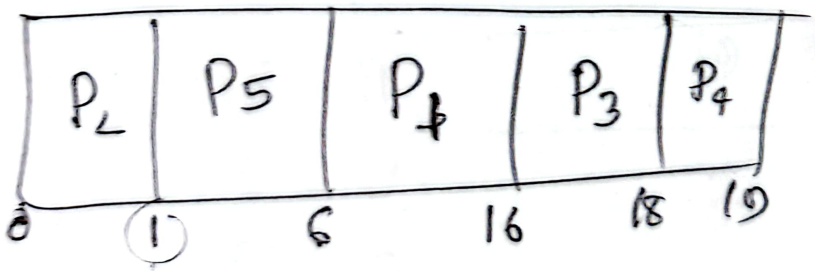
process	Burst	Priority
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2



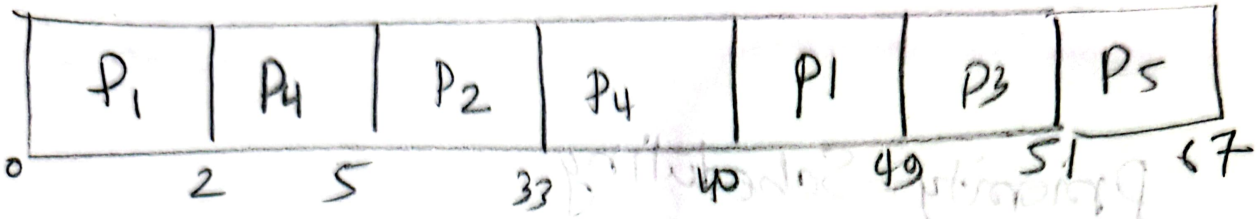
lab - 5

Priority Scheduling

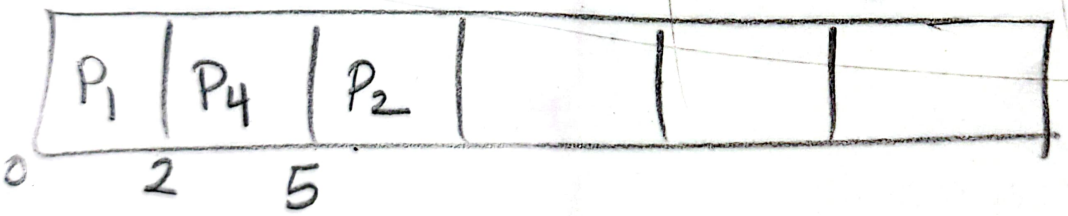
process	Burst	Priority
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2



Priority Scheduling Problem



	Burst	Arrival	Priority
P_1	9	0	2
P_2	28	5	0
P_3	2	12	3
P_4	10	2	1
P_5	16	9	4



P_1 = 9

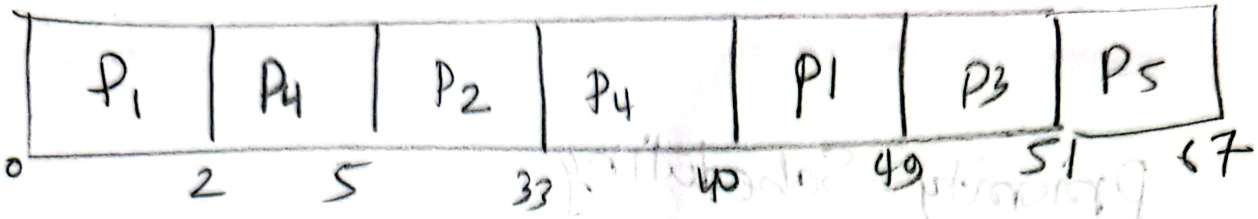
P_2 = 28

P_3 = 2

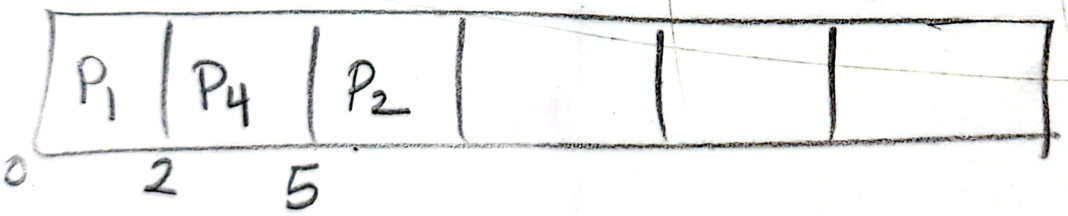
P_4 = 10

P_5 = 16

Priority Scheduling Problem



	Burst	Arrival	Priority
P_1	9	0	2
P_2	28	5	0
P_3	2	12	3
P_4	10	2	1
P_5	16	9	4



$P_1 = 9$

$P_2 = 28$

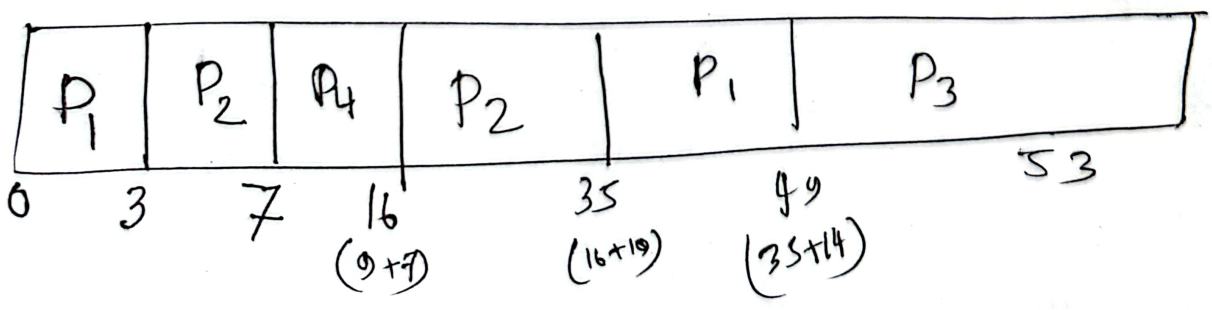
$P_3 = 2$

$P_4 = 10$

$P_5 = 16$

	Completion − arrival = Turnaround	Waiting = Turnaround − Burst	Response time = Start − Arrival
P_1	49 − 0 = 49	49 − 11 = 38	0 − 0 = 0
P_2	33 − 5 = 28	28 − 28 = 0	5 − 5 = 0
P_3	51 − 12 = 39	39 − 2 = 37	49 − 12 = 37
P_4	40 − 2 = 38	38 − 10 = 28	33 − 2 = 31
P_5	67 − 9 = 58	58 − 16 = 42	67 − 9 = 58
	= 42.4	2.0	15.8

PID	Burst	Arrival	priority
P_1	17	0	2
P_2	23	3	1
P_3	4	10	3
P_4	9	7	0



$P_1 = 17 - 3 = 14$

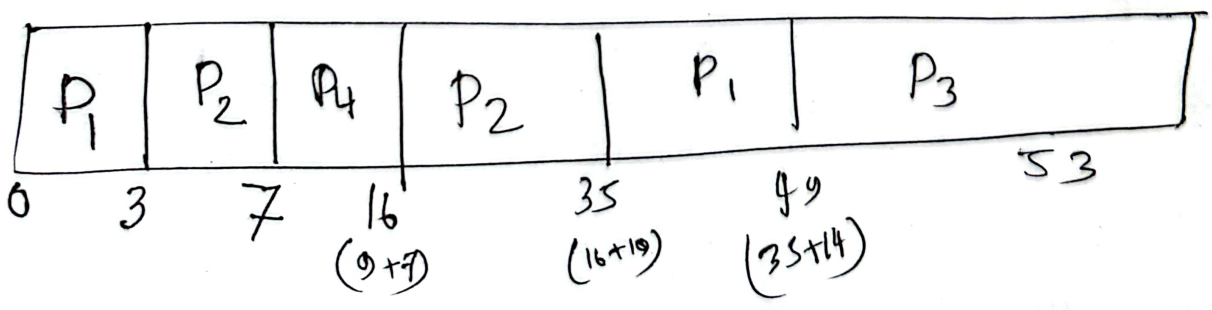
$P_2 = 23 - 4 = 19$

$P_3 = 4$

$P_4 = 9$

	Completion − arrival = Turnaround	Waiting = Turnaround − Burst	Response time = Start − Arrival
P_1	49 − 0 = 49	49 − 11 = 38	0 − 0 = 0
P_2	33 − 5 = 28	28 − 28 = 0	5 − 5 = 0
P_3	51 − 12 = 39	39 − 2 = 37	49 − 12 = 37
P_4	40 − 2 = 38	38 − 10 = 28	33 − 2 = 31
P_5	67 − 9 = 58	58 − 16 = 42	67 − 9 = 58
	= 42.4	2.0	15.8

PID	Burst	Arrival	priority
P_1	17	0	2
P_2	23	3	1
P_3	4	10	3
P_4	9	7	0



$P_1 = 17 - 3 = 14$

$P_2 = 23 - 4 = 19$

$P_3 = 4$

$P_4 = 9$

	Turnaround = completion − Arrival	Waiting time = Turnaround − Burst	Response Time = Start − Arrival
P_1	49 − 0 = 49	49 − 17 = 32	0 − 0 = 0
P_2	35 − 3 = 32	32 − 23 = 9	3 − 3 = 0
P_3	53 − 19 = 34	34 − 4 = 30	49 − 19 = 30
P_4	16 − 7 = 9	9 − 9 = 0	7 − 7 = 0

	Turnaround = completion − Arrival	Waiting time = Turnaround − Burst	Response Time = Start − Arrival
P_1	49 − 0 = 49	49 − 17 = 32	0 − 0 = 0
P_2	35 − 3 = 32	32 − 23 = 9	3 − 3 = 0
P_3	53 − 19 = 34	34 − 4 = 30	49 − 19 = 30
P_4	16 − 7 = 9	9 − 9 = 0	7 − 7 = 0

Problem of this algorithm:

Starvation ⇒ low priority processes may never execute

~~Solution ⇒~~ Aging ⇒ as time progresses increase the priority of the process

Round Robin (RR) Algorithm

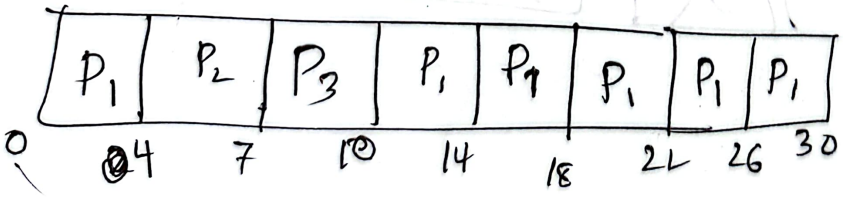
Round Robin ও priority মাত্রের মধ্যে (OS)

Process	Burst
P_1	24
P_2	3
P_3	3

Given

time_quantum = 4

4 sec এর জন্য
প্রত্যেক run করবে
সমানভাবে



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Round Robin (RR) Algorithm

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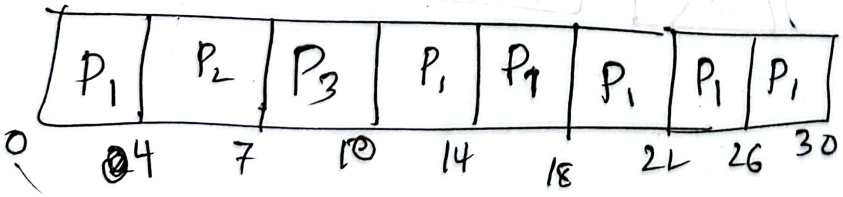
Given

time_quantum = 4

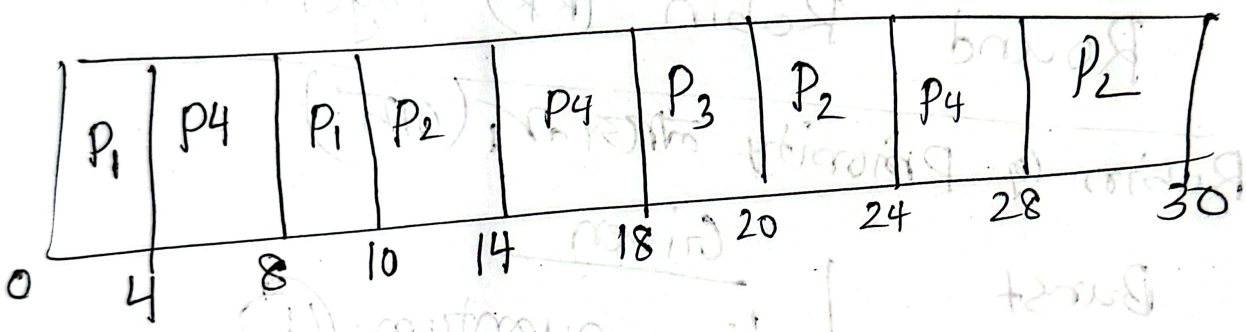
4 sec এর জন্য
প্রত্যেক run করবে
সমানভাবে



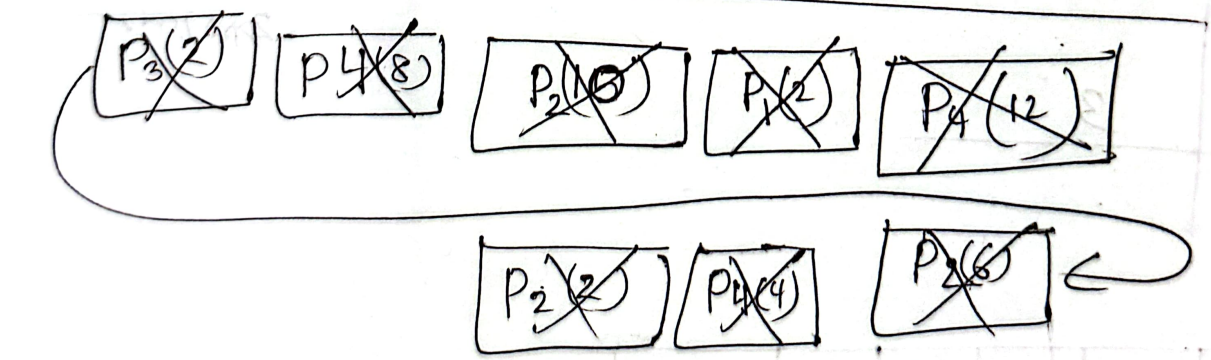
PID	Burst	Arrival
P_1	6	0
P_2	10	5
P_3	2	10
P_4	12	2

Given

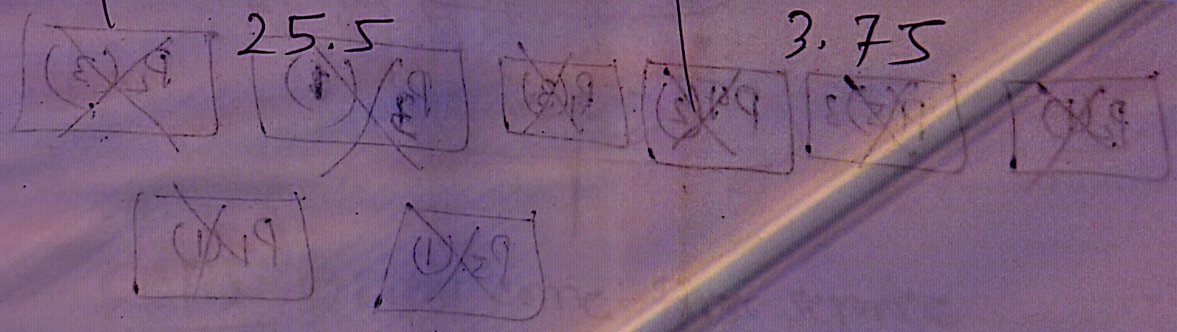
Time_quantum = 4

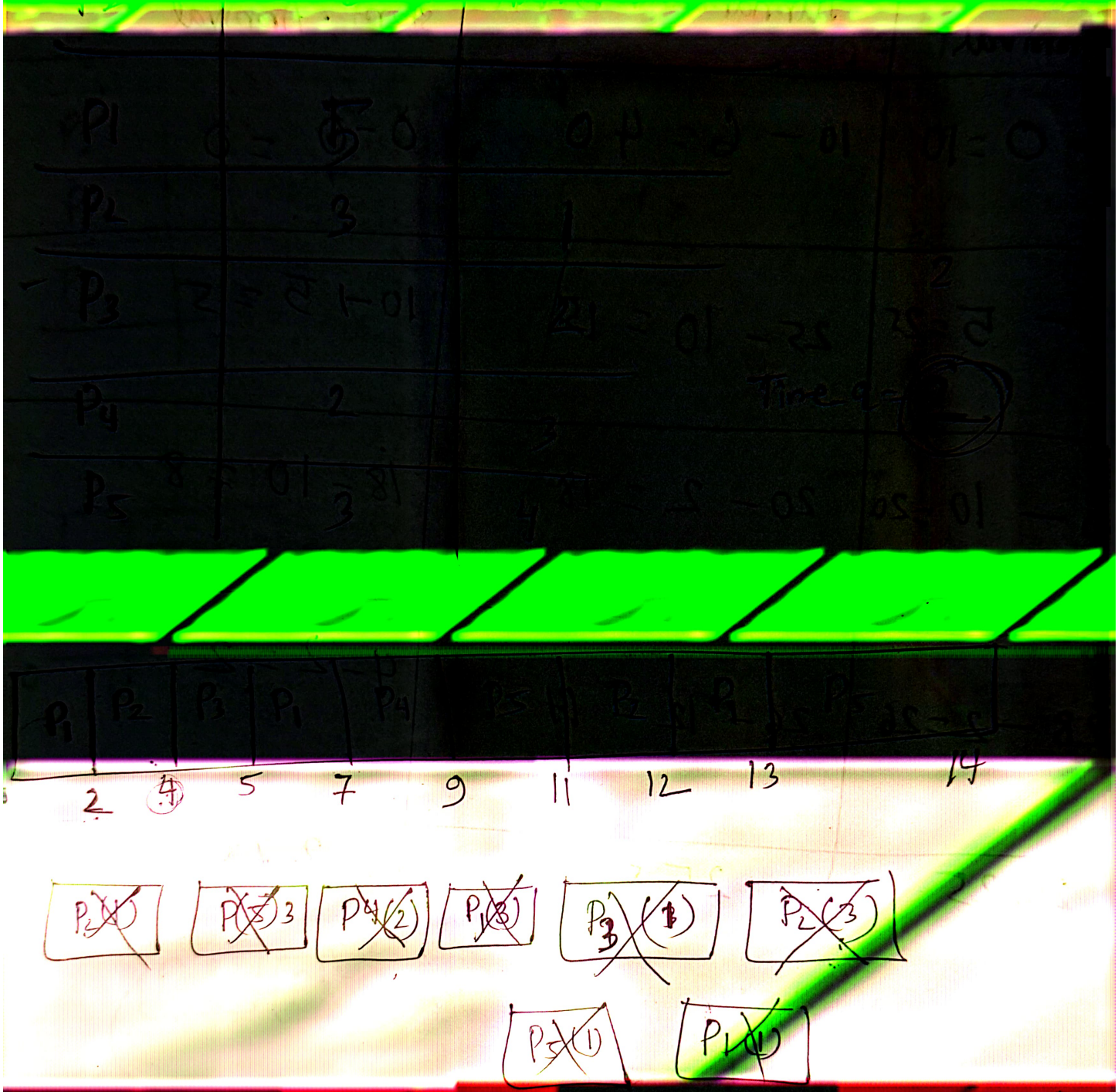


queue table



Turnaround: Completion − Arrival	Waiting: Turnaround − Burst	Response Time = start − Arrival
10 − 0 = 10	10 − 6 = 4	0 − 0 = 0
30 − 5 = 25	25 − 10 = 15	15 − 5 = 5
28 − 2 = 26	26 − 12 = 14	4 − 2 = 2
20.25	25.5	3.75



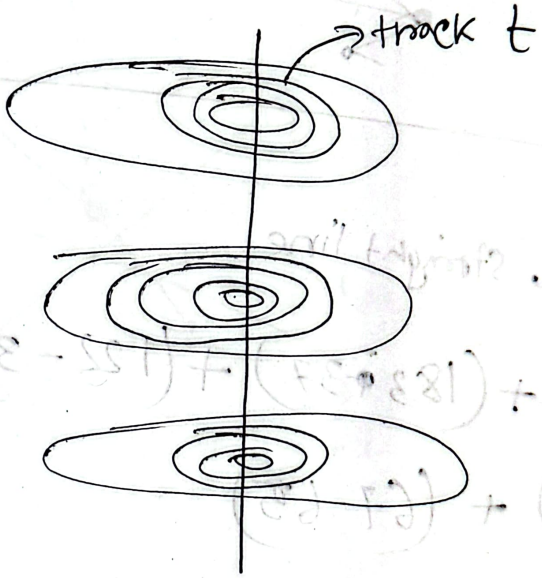


lab 6

```
Void solveFCFS(){
    int n;
    cin>>n;
```

Moving-head Disk Mechanism (Theory)

platter ⇒ layer

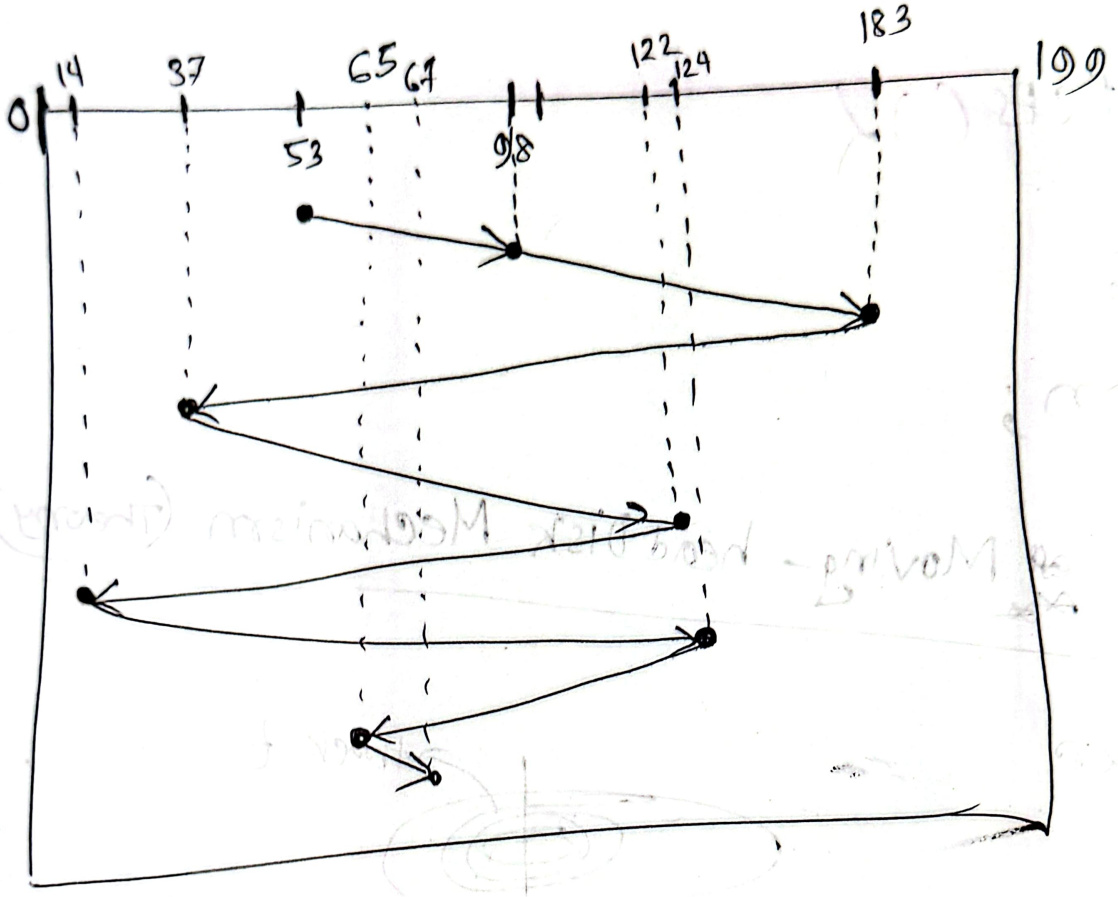


FCFS ⇒ First come first service

Given ⇒ queue: 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53

⇒

Diagram

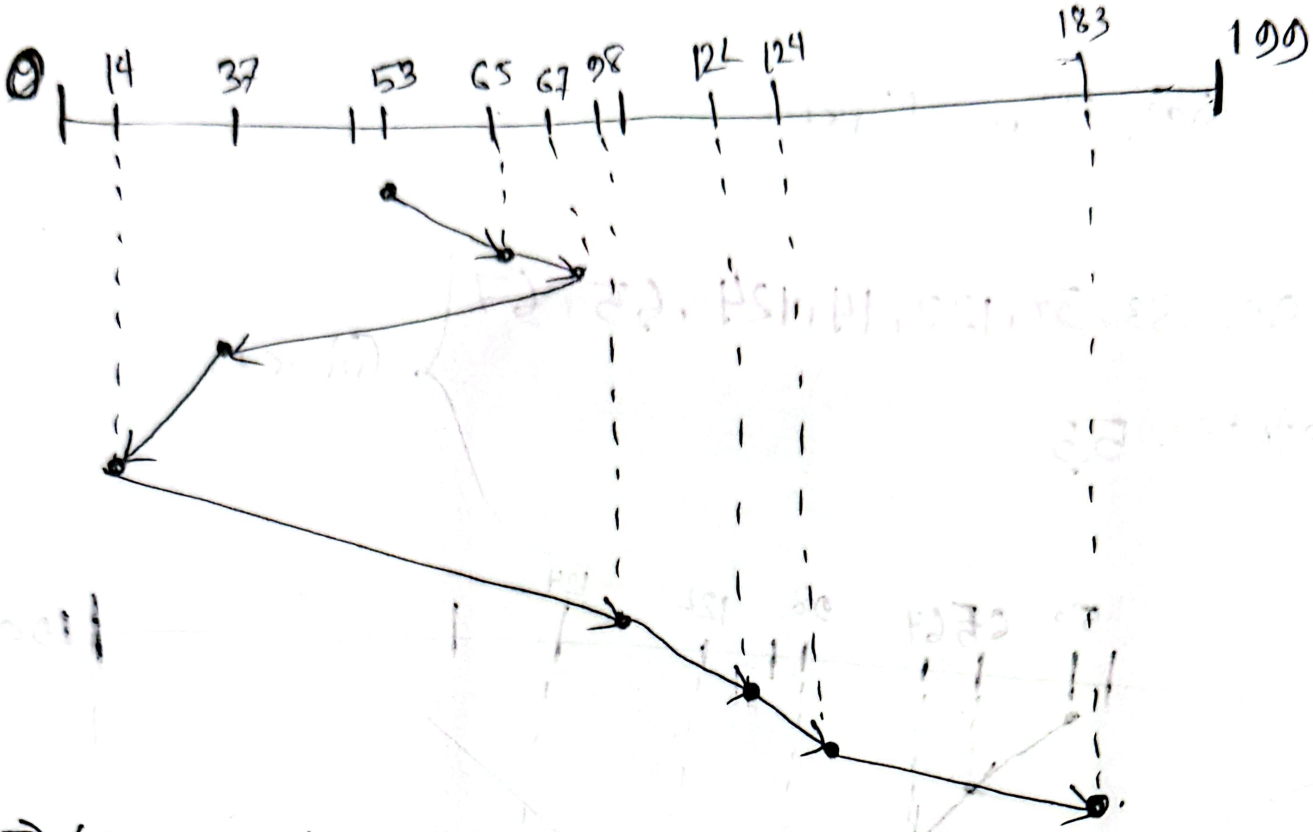


length of the straight line:

$$(183-53)+(183-37)+(122-37)+(122-14)+$$

$$(124-65)+(67-65)$$

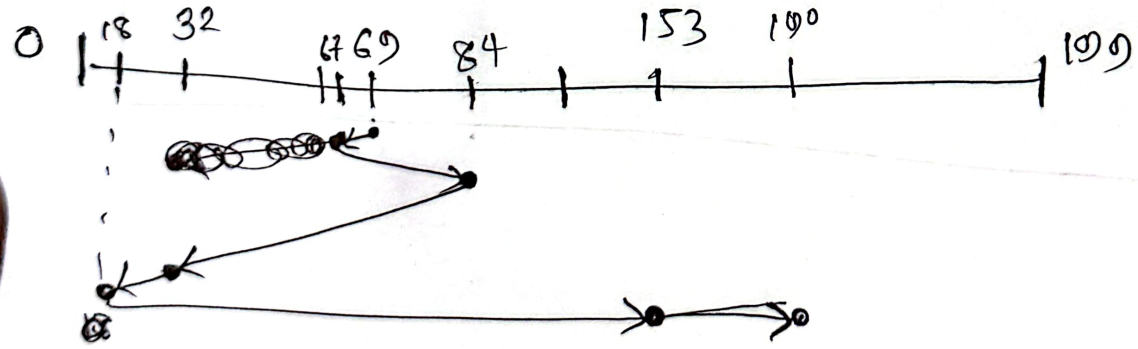
$$=$$



$\Rightarrow (67-53) + (67-14) + (183-14)$

④. 84, 153, 32, 67, 100, 18

⇒ 69



$= (69-67) + (84-67) + (84-18) + (100-18)$

$=$

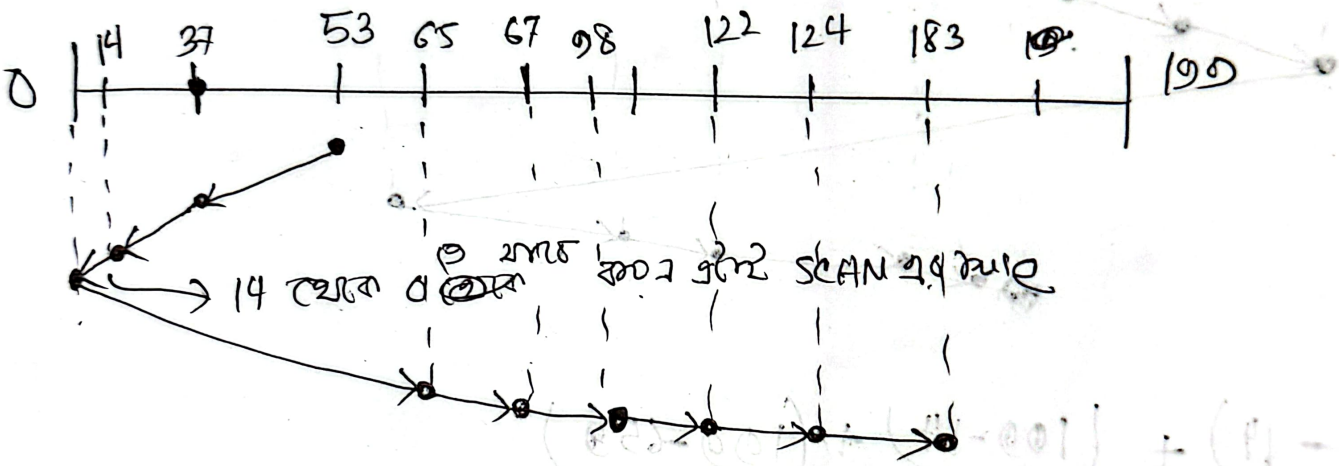
SCAN / elevator algorithm.

⇒ SCAN এ Direction preference দিতে হয়

queue ⇒ 98, 183, 37, 122, 14, 124, 65, 67 } Given

Head ⇒ 53

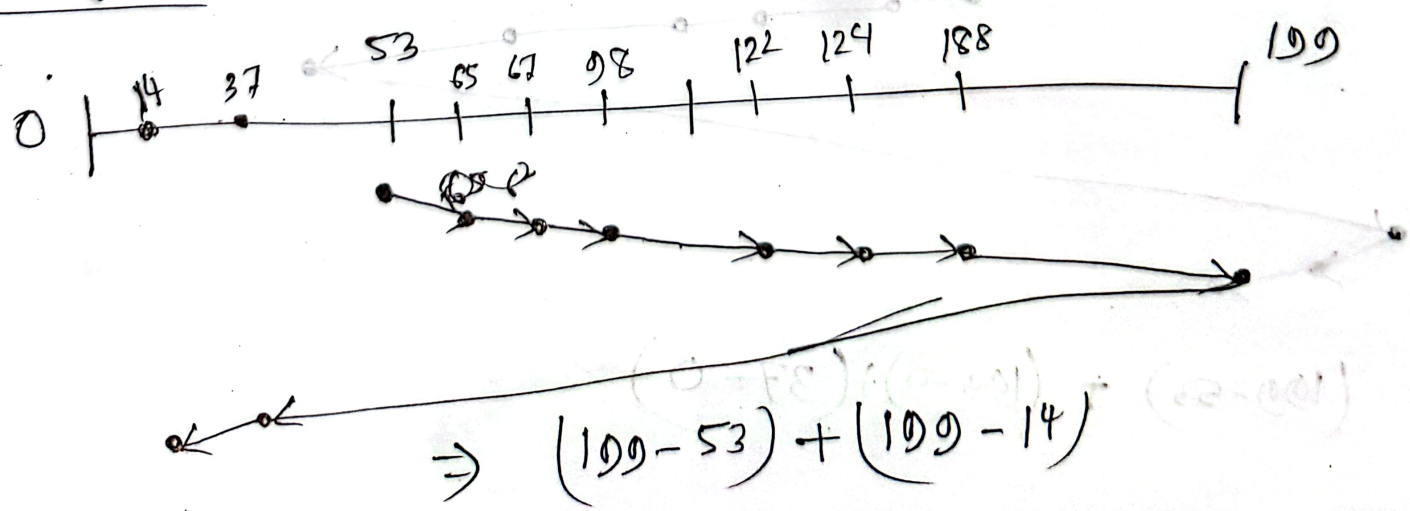
Direction ⇒ lower



$(53-0)+(183-0)$

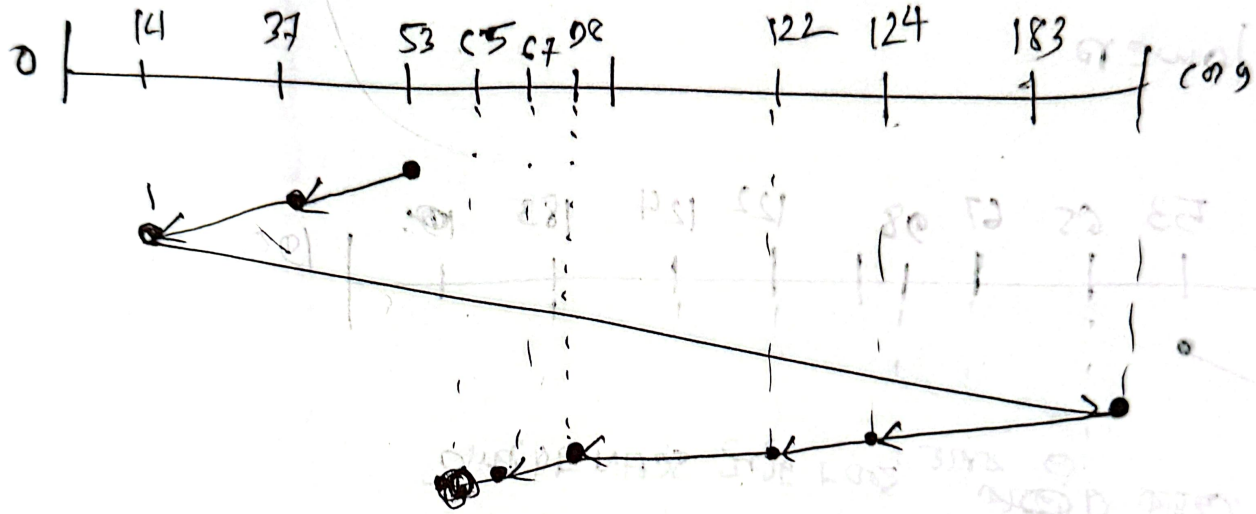
$= 236$

for higher!



C-Scan

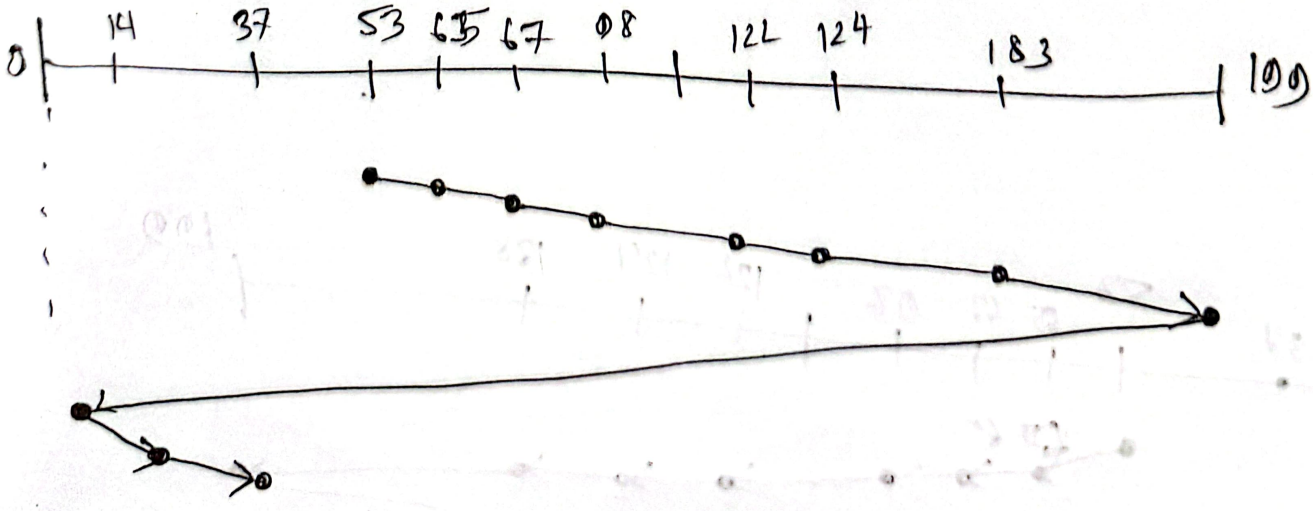
almost scan এর মতো But এখানে দুই end point এ যেতে হয়.



$(53 - 14) + (199 - 14) + (199 - 65)$

$\Rightarrow$

higher

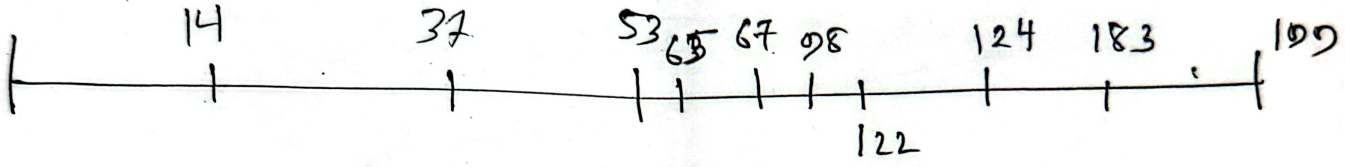


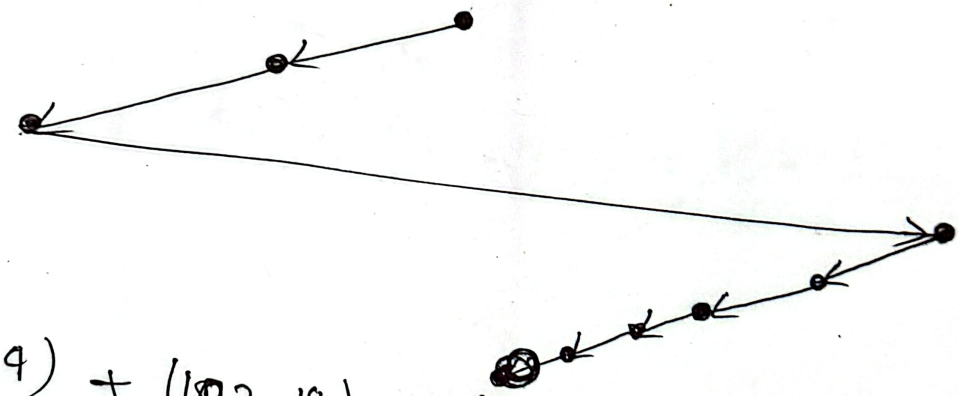
$(199 - 53) + (199 - 0) + (37 - 0)$

C-look

preference ঘোড়া থাকবে। এখানে end point এ যাওয়া লাগেনা C-scan এর মতো। এখানে যখন Jump করবে তখন সর্বনিম্নে যেই এবং সর্বোচ্চে যেতে হবে।

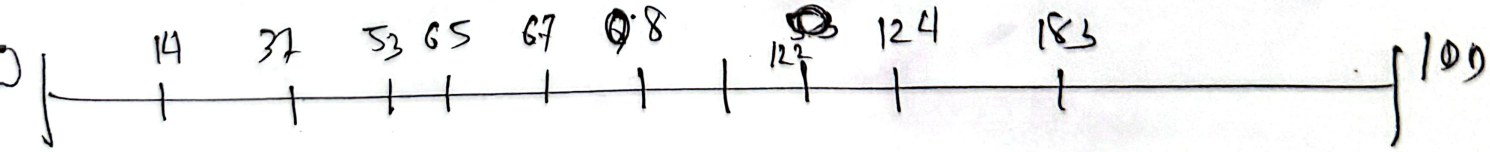
lower

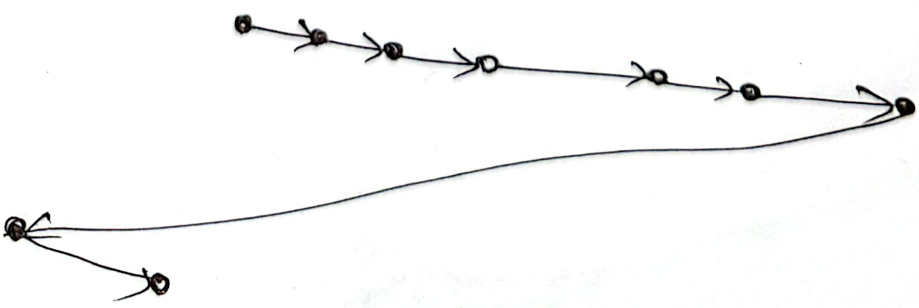




$(53-14) + (183-14) + (183-65)$

higher



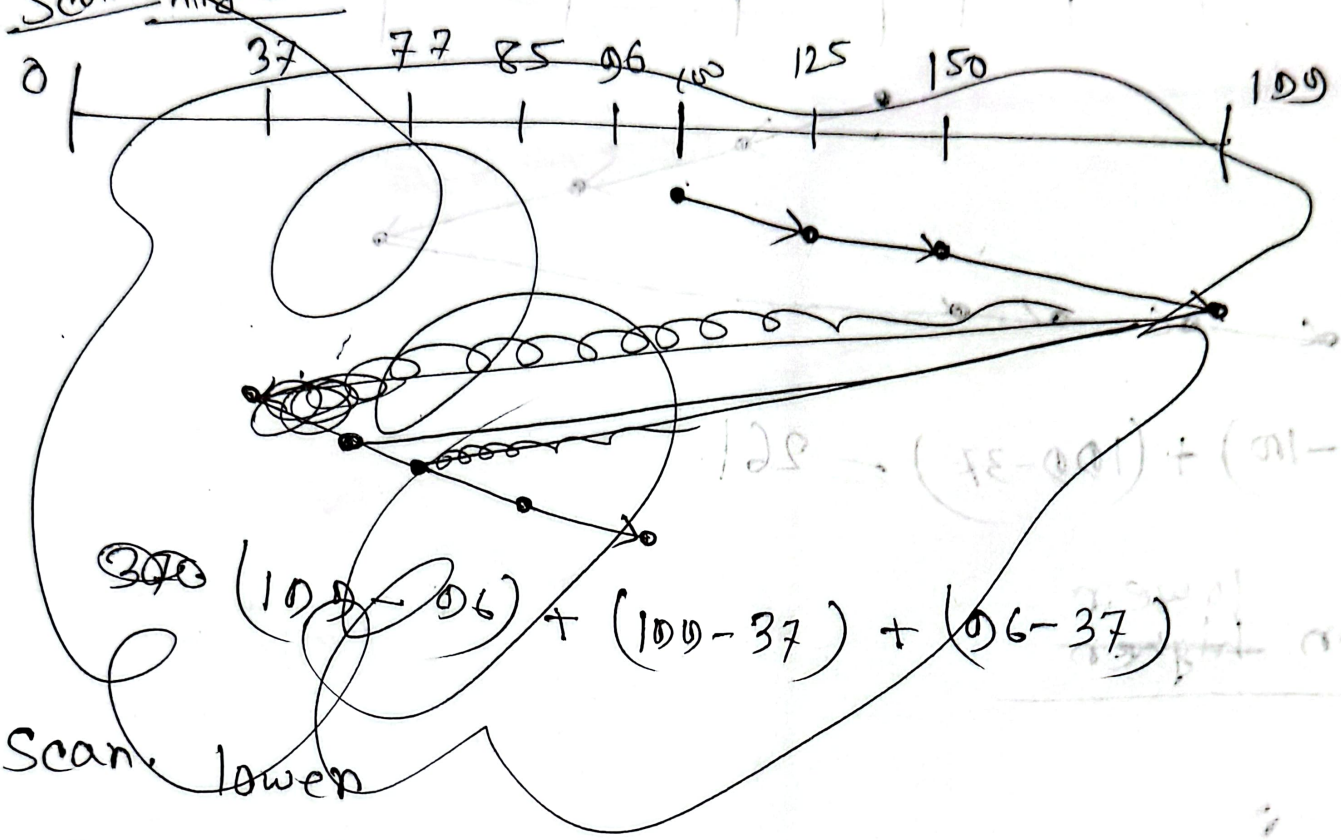


$(183-53) + (183-0) + (37-14)$

(b). 85, 37, 150, 77, 125, 96

Head = 100

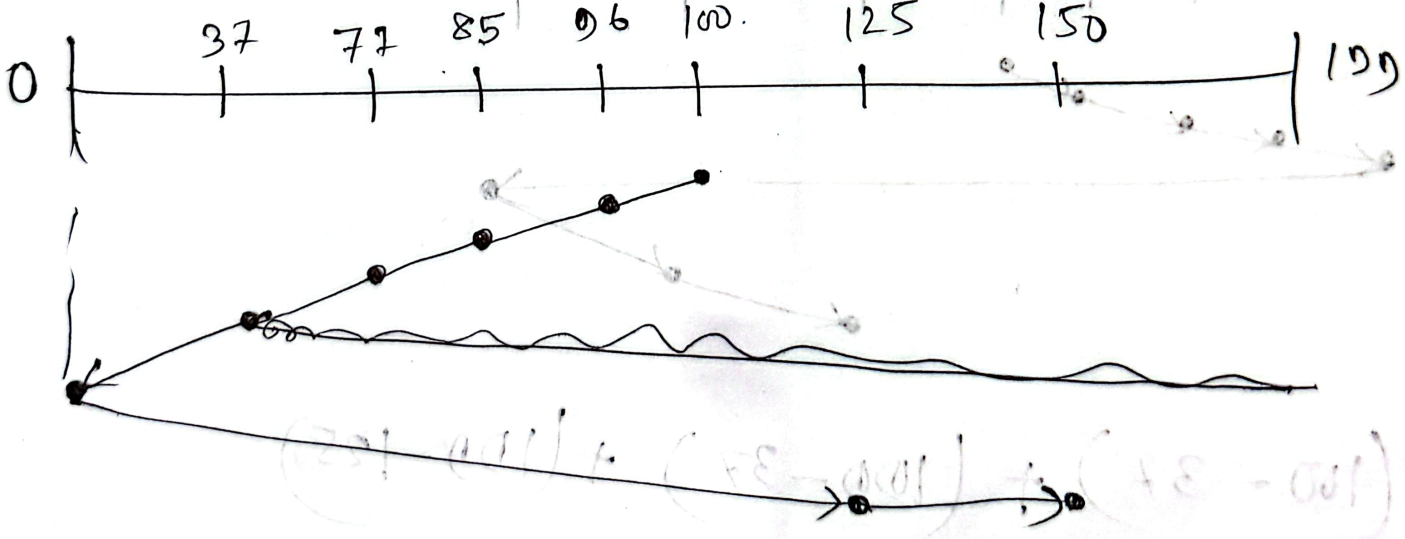
Scan higher:



(199 − 96) + (199 − 37) + (96 − 37)

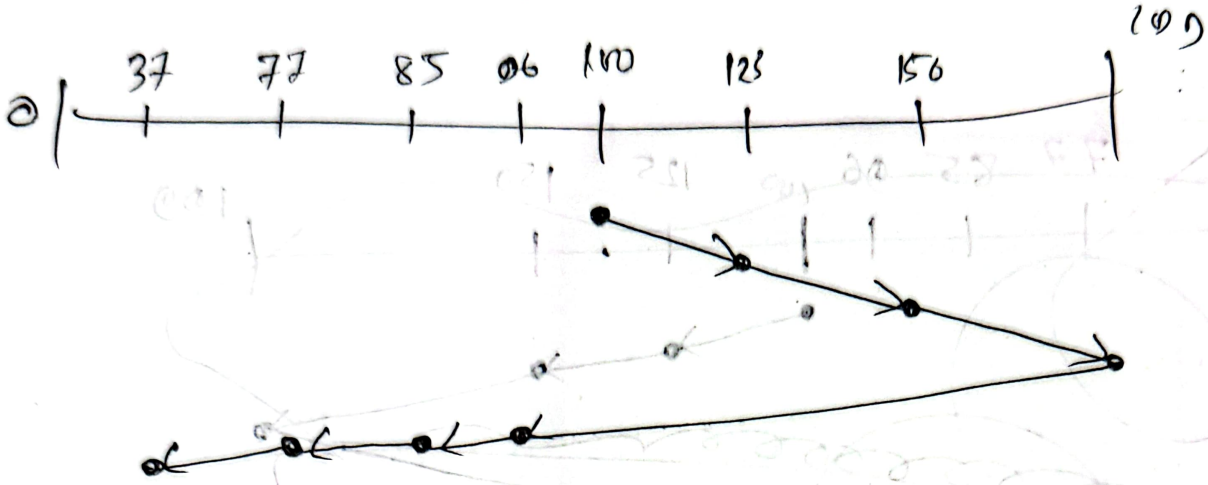
Scan lower

Scan lower



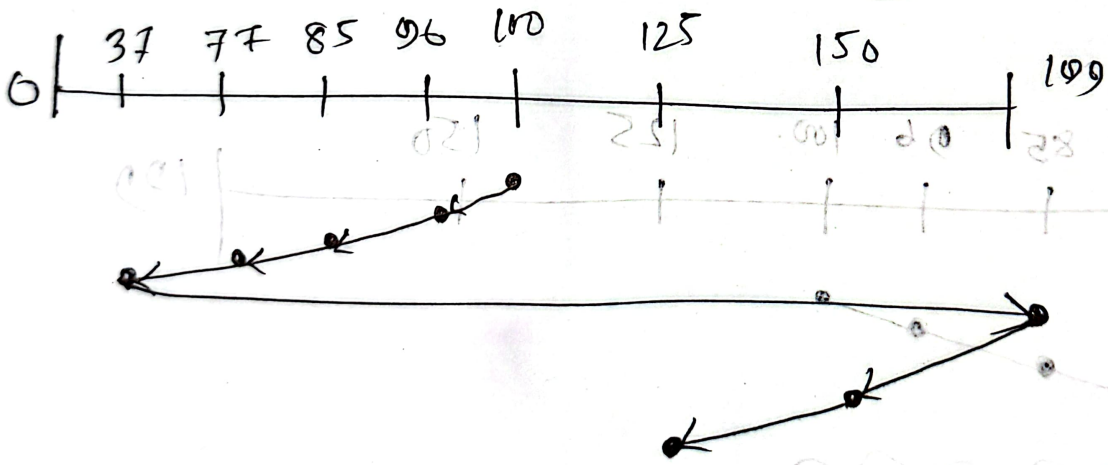
(100 − 0) + (150 − 0) = 250

scan higher



$(199-100)+(100-37)=261$

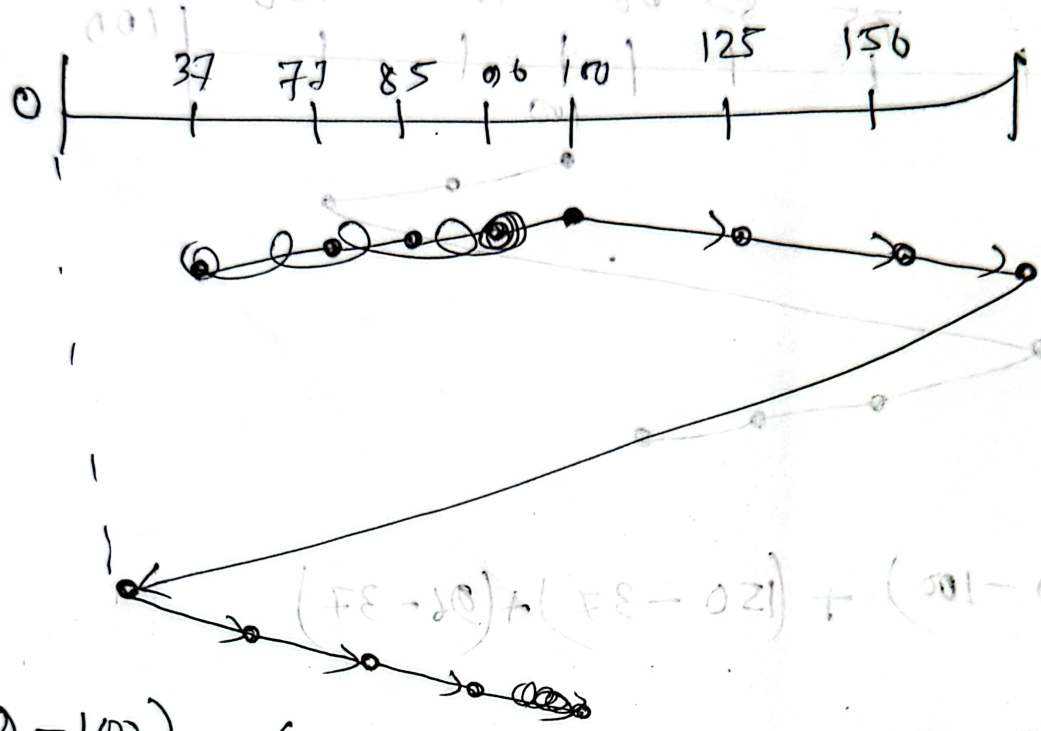
scan ~~higher~~ lower



$(100-37)+(199-37)+(199-125)$

$=299$

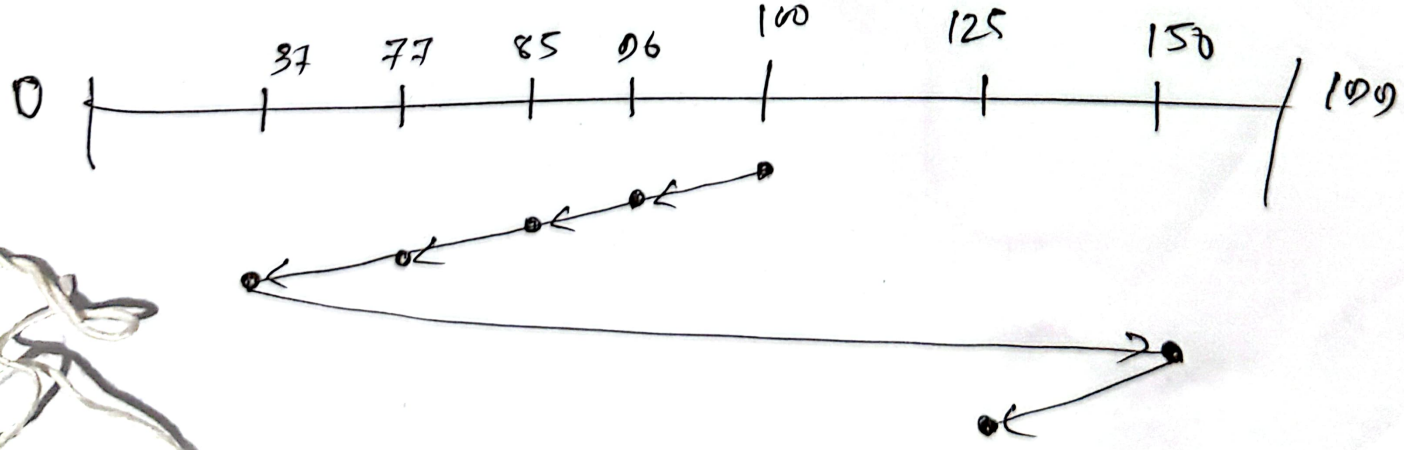
higher
C-scan ~~tower~~



$(199-100) + (199-0) + (96-0) = 394$

C-look

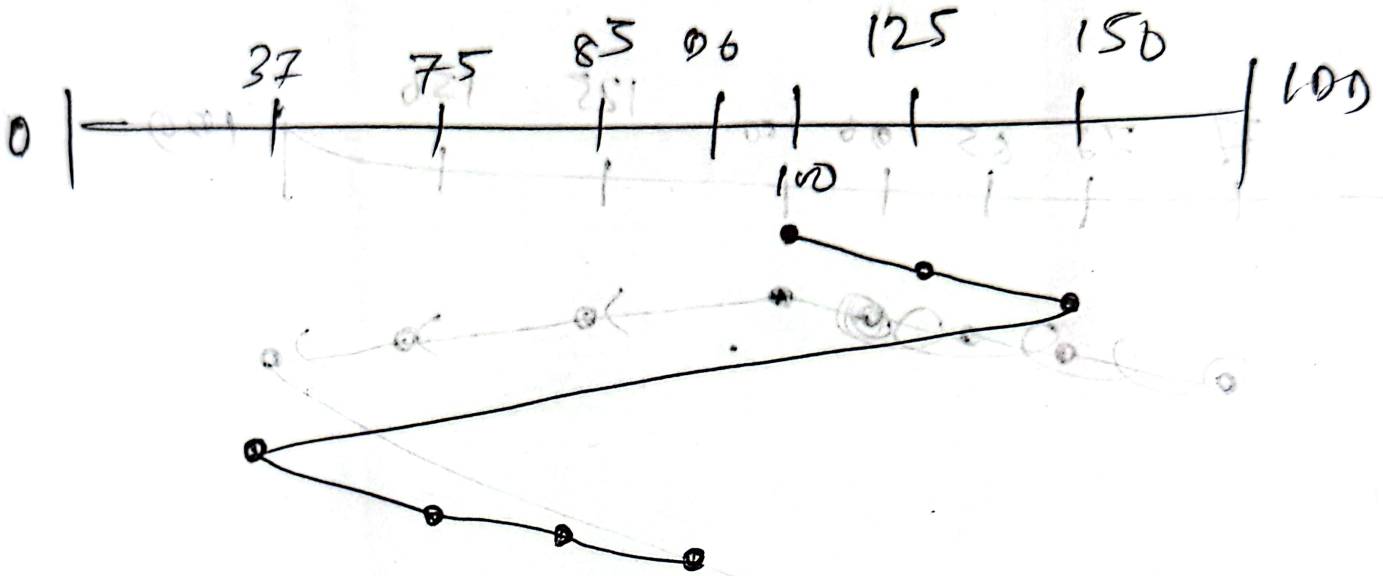
lower



$(100-37) + (150-37) + (150-125)$

$= 201$

higher



$= (150-100) + (150-37) + (96-37)$

$= 222$